

that destroy phagocytes before phagocytosis can occur. *Streptococcus pyogenes* releases a protease that cleaves the C5a complement factor and thus inhibits complement's ability to attract phagocytes to the infected area.

Survival Inside Phagocytic Cells

Some bacteria have evolved the ability to survive inside neutrophils, monocytes, and macrophages. Such pathogens are very pathogenic because they are impervious to a most important host protective mechanism. One evading mechanism is to escape from the phagosome before it merges with the lysosome, as seen with *Listeria monocytogenes*, *Shigella*, and *Rickettsia*. Another approach is to resist the toxic products released into the phagolysosome after fusion occurs. A good example of a bacterium that is resistant to the lysosomal enzymes is *Mycobacterium tuberculosis*, probably at least partly because of its waxy external layer. Still other bacteria prevent fusion of phagosomes with lysosomes (*Chlamydia*). Phagocytosis (pp. 718–20)

Evading the Specific Immune Response

To evade the specific immune response, some bacteria (*Streptococcus pyogenes*) produce capsules that are not antigenic since they resemble host tissue components. *N. gonorrhoeae* can also evade the specific immune response by two mechanisms: (1) it makes genetic variations in its pili (phase variation) so that specific antibodies are useless against the new pili and adherence to host tissue occurs, and (2) it produces IgA proteases that destroy secretory IgA and allow adherence. Finally, some bacteria produce proteins (such as staphylococcal protein A and protein G of *Streptococcus pyogenes*) that interfere with antibody-mediated opsonization (see figure 31.15) by binding to the Fc portion of immunoglobulins.

1. What are some mechanisms viruses use to evade host defenses?
2. How do bacteria evade each of the following host defenses: the complement system, phagocytosis, and the specific immune response?

Summary

1. Parasitism is a type of symbiosis between two species in which the smaller organism is physiologically dependent on the larger one, termed the host. The parasitic organism usually harms its host in some way.
2. An infection is the colonization of the host by a parasitic organism. An infectious disease is the result of the interaction (figure 34.1) between the parasitic organism and its host, causing the host to change from a state of health to a diseased state. Any organism that produces such a disease is a pathogen.
3. Pathogenicity refers to the quality or ability of an organism to produce pathological changes or disease. Virulence refers to the degree or intensity of pathogenicity of an organism and is measured experimentally by the LD₅₀ or ID₅₀ (figure 34.2).
4. The fundamental process of viral infection is the expression of the viral replicative cycle in a host cell. To produce disease a virus must enter a host; come into contact with susceptible cells; reproduce; spread to adjacent cells; cause cellular injury; engender a host immune response; be either cleared from the body of the host, establish a persistent infection, or kill the host; and be shed back into the environment.
5. Pathogens or their products can be transmitted to a host by either direct or indirect means. Transmissibility is the initial requisite in the establishment of an infectious disease.
6. Special adherence factors (table 34.3) allow pathogens to bind to specific receptor sites on host cells and colonize the host (figure 34.3).
7. Pathogens can enter host cells by both active and passive mechanisms. Once inside, they can produce specific products and/or enzymes that promote dissemination throughout the body of the host. These are termed virulence factors (table 34.4).
8. The pathogen generally is found in the area of the host's body that provides the most favorable conditions for its growth and multiplication.
9. In bacteria, one important result of the conservation of chromosomal genes is that bacteria are clonal. For most pathogenic bacteria, there are only a few clonal types that exist in the environment.
10. During coevolution with human hosts, some pathogenic bacteria have evolved complex signal transduction pathways to regulate the genes necessary for virulence.
11. Many bacteria are pathogenic because they have large segments of DNA called pathogenicity islands that carry genes responsible for virulence.
12. Intoxications are diseases that result from the entrance of a specific toxin into a host. The toxin can induce the disease in the absence of the toxin-producing organism. Toxins produced by pathogens can be divided into two main categories: exotoxins and endotoxins (table 34.5).
13. Exotoxins are soluble, heat-labile, potent, toxic proteins produced by the pathogen as a result of its normal metabolism. They have very specific effects and can be categorized as neurotoxins, cytotoxins, or enterotoxins. Most exotoxins conform to the AB model in which the A subunit or fragment is enzymatic and the B subunit or fragment, the binding portion (table 34.6). Several mechanisms exist by which the A component enters target cells (figure 34.5).
14. Exotoxins can be divided into four types: (a) the AB toxins, (b) specific host site toxins (neurotoxins, enterotoxins, cytotoxins), (c) toxins that disrupt plasma membranes of host cells (leukocidins, hemolysins, and phospholipases), and (d) superantigens.
15. Bacterial exotoxins cause disease in a human host in three main ways: (a) ingestion of preformed exotoxin, (b) colonization of a mucosal surface followed by exotoxin production, and (c) colonization of a wound followed by local exotoxin production.
16. Endotoxins are heat-stable, toxic substances that are part of the cell wall lipopolysaccharide of some gram-negative bacteria. Most endotoxins function by initially activating Hageman Factor, which in turn activates one to four humoral systems. These include the intrinsic blood clotting cascade, complement activation, fibrinolytic system, and kininogen system. Endotoxins also stimulate macrophages to release cytokines such as IL-1, IL-6, and TNF- α .
17. During the course of microbe and human evolution, some pathogens have evolved ways for escaping host defenses. Viruses have mechanisms that either suppress or evade the host's immune response. Bacteria have evolved mechanisms to evade the complement system, phagocytosis, and the specific immune response.

Key Terms

AB toxins 797
adhesin 792
alpha hemolysis 797
antitoxin 796
bacteremia 793
beta hemolysis 797
colonization 792
cytotoxin 797
ectoparasite 788
endogenous pyrogen 801
endoparasite 789
endotoxin 799
endotoxin unit (E.U.) 800
enterotoxin 797
exotoxin 794
final host 789
fomite 792
hemolysin 797
host 788
immunopathology 790

infection 789
infectious disease 789
infectious dose 50 (ID₅₀) 790
infectivity 790
intermediate host 789
intoxication 794
invasiveness 790
lethal dose 50 (LD₅₀) 790
leukocidin 797
LPS-binding protein 801
membrane-disrupting exotoxin 797
neurotoxin 797
opportunistic pathogen 789
parasite 788
parasitic organism 788
parasitism 788
pathogen 789
pathogenicity 789
pathogenic potential 790
pathogenicity island 794

phospholipase 797
primary (frank) pathogen 789
reservoir 791
reservoir host 789
septicemia 793
serum resistance 801
streptolysin-O (SLO) 797
streptolysin-S (SLS) 797
toxemia 794
toxigenicity 790
toxin 794
toxoid 796
transfer host 789
tropism 791
type III secretion system 794
vector 791
viremia 791
virulence 790
virulence factor 792

Questions for Thought and Review

1. Why does a parasitic organism not have to be a parasite?
2. In general, infectious diseases that are commonly fatal are newly evolved relationships between the parasitic organism and the host. Why is this so?
3. What does an organism require to be parasitic?
4. What are some bacterial determinants that provide the organism with the ability to colonize and invade the host?
5. What is the significance of the clonal nature of bacterial pathogens?
6. How do some bacteria regulate their virulence factors?
7. Describe a pathogenicity island.
8. What are four types of exotoxins based on their structural and physiological activities?
9. What is the difference between the general properties of endotoxins and exotoxins?
10. How do some viruses evade the defenses of a human host? How do some bacteria evade the defenses of a human host?

Critical Thinking Questions

1. Explain the observation that different pathogens infect different parts of the host.
2. Intracellular bacterial infections present a particular difficulty for the host. Why is it harder to defend against these infections than against viral infections and extracellular bacterial infections?

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CHAPTER 35

Antimicrobial Chemotherapy



Many antimicrobial medications are available to combat infections.

Outline

- | | | | |
|------|--|------|--|
| 35.1 | The Development of Chemotherapy 806 | 35.6 | Antibacterial Drugs 812 |
| 35.2 | General Characteristics of Antimicrobial Drugs 807 | | Sulfonamides or Sulfa Drugs 812 |
| 35.3 | Determining the Level of Antimicrobial Activity 809 | | Quinolones 813 |
| | Dilution Susceptibility Tests 809 | | Penicillins 813 |
| | Disk Diffusion Tests 809 | | Cephalosporins 814 |
| | Measurement of Drug Concentrations in the Blood 809 | | The Tetracyclines 815 |
| 35.4 | Mechanisms of Action of Antimicrobial Agents 810 | | Aminoglycoside Antibiotics 816 |
| 35.5 | Factors Influencing the Effectiveness of Antimicrobial Drugs 812 | | Erythromycin and Other Macrolides 817 |
| | | | Vancomycin and Teicoplanin 817 |
| | | | Chloramphenicol 817 |
| | | 35.7 | Drug Resistance 818 |
| | | | Mechanisms of Drug Resistance 818 |
| | | | The Origin and Transmission of Drug Resistance 819 |
| | | 35.8 | Antifungal Drugs 820 |
| | | 35.9 | Antiviral Drugs 821 |

Concepts

1. Many diseases are treated with chemotherapeutic agents, such as antibiotics, that inhibit or kill the pathogen while harming the host as little as possible.
2. Ideally, antimicrobial agents disrupt microbial processes or structures that differ from those of the host. They may damage pathogens by hampering cell wall synthesis, inhibiting microbial protein and nucleic acid synthesis, disrupting microbial membrane structure and function, or blocking metabolic pathways through inhibition of key enzymes.
3. The effectiveness of chemotherapeutic agents depends on many factors: the route of administration and location of the infection, the presence of interfering substances, the concentration of the drug in the body, the nature of the pathogen, the presence of drug allergies, and the resistance of microorganisms to the drug.
4. The increasing number and variety of drug-resistant pathogens is a serious public health problem.
5. Although antibacterial chemotherapy is more advanced, drugs for the treatment of fungal and viral infections are also becoming increasingly available.

It was the knowledge of the great abundance and wide distribution of actinomycetes, which dated back nearly three decades, and the recognition of the marked activity of this group of organisms against other organisms that led me in 1939 to undertake a systematic study of their ability to produce antibiotics.

—Selman A. Waksman

The control of microorganisms is critical for the prevention and treatment of disease. Chapter 7 is concerned principally with the chemical and physical agents used to treat inanimate objects in order to destroy microorganisms or inhibit their growth. In addition, the use of antiseptics is briefly reviewed. Microorganisms also grow on and within other organisms, and microbial colonization can lead to disease, disability, and death. Thus the control or destruction of microorganisms residing within the bodies of humans and other animals is of great importance.

When disinfecting or sterilizing an inanimate object, one naturally must use procedures that do not damage the object itself. The same is true for the treatment of living hosts. The most successful drugs interfere with processes that differ between the pathogen and host, and seriously damage the target microorganism while harming its host as little as possible. This chapter introduces the principles of chemotherapy and briefly reviews the characteristics of selected antibacterial, antifungal, and antiviral drugs.

Modern medicine is dependent on **chemotherapeutic agents**, chemical agents that are used to treat disease. Chemotherapeutic agents destroy pathogenic microorganisms or inhibit their growth at concentrations low enough to avoid undesirable damage to the host. Most of these agents are **antibiotics** [Greek *anti*, against, and *bios*, life], microbial products or their derivatives that can kill susceptible microorganisms or inhibit their growth. Drugs such as the sulfonamides are sometimes called antibiotics although they are synthetic chemotherapeutic agents, not microbially synthesized.

35.1 The Development of Chemotherapy

The modern era of chemotherapy began with the work of the German physician Paul Ehrlich (1854–1915). Ehrlich reasoned that a chemical with selective toxicity that would kill pathogens and not human cells might be effective in treating disease. He hoped to find a toxic dye molecule, a “magic bullet,” that would specifically bind to pathogens and destroy them; therefore he began experimenting with dyes. By 1904 Ehrlich found that the dye trypan red was active against the trypanosome that causes African sleeping sickness (see figure 27.3) and could be used therapeutically. Subsequently Ehrlich and a young Japanese scientist named Sahachiro Hata tested a variety of arsenicals on syphilis-infected rabbits and found that compound number 606, arsphenamine, was active against the syphilis spirochete (see figure 21.15b). Arsphenamine was made available in 1910 under the trade name Salvarsan. Ehrlich’s successes in the chemotherapy of sleeping sickness and syphilis established his concept of selective toxicity and led to the testing of hundreds of compounds for their therapeutic potential.

In 1927 the German chemical industry giant, I. G. Farbenindustrie, began a long-term search for chemotherapeutic agents under the direction of Gerhard Domagk. The company provided vast numbers of dyes and other chemicals that Domagk tested for activity against pathogenic bacteria and for toxicity in animals. During this screening program Domagk discovered that Prontosil Red, a new dye for staining leather, was nontoxic for animals and completely protected mice against pathogenic streptococci and staphylococci. These results were published in 1935, and in the same year the French scientists Jacques and Therese Trefouel showed that Prontosil Red was converted in the body to sulfanilamide, the true active factor. Domagk had actually discovered sulfonamides or sulfa drugs and for this discovery he received the Nobel Prize in 1939.

The story of the discovery and development of penicillin, the first antibiotic to be used therapeutically, is complex and fascinating. Although penicillin was actually discovered in 1896 by a 21-year-old French medical student named Ernest Duchesne, his work was forgotten, and penicillin was rediscovered and brought to the attention of scientists by the Scottish physician Alexander Fleming. Fleming had been interested in finding something that would kill pathogens ever since working on wound infections during the First World War. One day in September 1928, a *Penicillium notatum* spore accidentally landed on the surface of an exposed petri dish before it had been inoculated with staphylococci, and a new medical era was born. Although the precise events are still unclear, Ronald Hare has suggested that Fleming left the contaminated plate on a laboratory bench while he was on vacation. Because the first few days of the vacation were cool, the fungus grew more rapidly than the bacteria and produced penicillin. When the weather then turned warm, the bacteria began to grow and were lysed. On his return, Fleming noticed that a *Penicillium* colony was growing at one edge and that the staphylococci surrounding it had been destroyed. Rather than discarding the contaminated plate, he correctly deduced that his mold contaminant was producing a diffusible substance lethal to staphylococci. He began efforts to characterize what he called penicillin. He found that broth from a *Penicillium* culture contained peni-

Box 35.1

The Use of Antibiotics in Microbiological Research

Although the use of antibiotics in the treatment of disease is emphasized in this chapter, it should be noted that antibiotics are extremely important research tools. For example, they aid the cultivation of viruses by preventing bacterial contamination. When eggs are inoculated with a virus sample, antibiotics often are included in the inoculum to maintain sterility. Usually a mixture of antibiotics (e.g., penicillin, amphotericin, and streptomycin) also is added to tissue cultures used for virus cultivation and other purposes.

Researchers often use antibiotics as instruments to dissect metabolic processes by inhibiting or blocking specific steps and observing the consequences. Although selective toxicity is critical when antibiotics are employed therapeutically, specific toxicity is more important in this context: the antibiotic must act by a specific and precisely understood mechanism. A clinically useful antimicrobial agent such as penicillin sometimes may be employed in research, but often an agent with specific toxicity and excellent research potential is too toxic for therapeutic use. The actinomycins, discovered in 1940 by Selman Waksman, are a case in point. They are so toxic to higher organisms that it was suggested they be used as rat poison. Today actinomycin D is a standard research tool

specifically used to block RNA synthesis. Other examples of antibiotics useful in research, with the process inhibited, are the following: chloramphenicol (bacterial protein synthesis), cycloserine (peptidoglycan synthesis), nalidixic acid and novobiocin (bacterial DNA synthesis), rifampin (bacterial RNA synthesis), cycloheximide (eucaryotic protein synthesis), daunomycin (fungal RNA synthesis), mitomycin C (DNA synthesis), polyoxin D (fungal cell wall chitin synthesis), and cerulenin (fatty acid synthesis).

In practice, the antibiotic is administered and changes in cell function are monitored. If one desired to study the dependence of bacterial flagella synthesis on RNA transcription, the flagella could be removed by high-speed mixing in a blender, followed by actinomycin D addition to the incubation mixture. The bacterial culture would then be observed for flagella regeneration in the absence of RNA synthesis. The results of such experiments must be interpreted with caution. Flagella synthesis may have been blocked because actinomycin D inhibited some other process, thus affecting flagella regeneration indirectly rather than simply inhibiting transcription of a gene required for flagella synthesis. Furthermore, not all microorganisms respond in the same way to a particular drug.

cillin and that the antibiotic could destroy several pathogenic bacteria. Unfortunately Fleming's next experiments convinced him that penicillin would not remain active in the body long enough after injection to destroy pathogens. After giving a talk on penicillin and publishing several papers on the subject between 1929 and 1931, he dropped the research.

In 1939 Howard Florey, a professor of pathology at Oxford University, was in the midst of testing the bactericidal activity of many substances, including lysozyme and the sulfonamides. After reading Fleming's paper on penicillin, one of Florey's coworkers, Ernst Chain, obtained the *Penicillium* culture from Fleming and set about culturing it and purifying penicillin. Florey and Chain were greatly aided in this by the biochemist Norman Heatley. Heatley devised the original assay, culture, and purification techniques needed to produce crude penicillin for further experimentation. When the purified penicillin was injected into mice infected with streptococci or staphylococci, practically all the mice survived. Florey and Chain's success was reported in 1940, and subsequent human trials were equally successful. Fleming, Florey, and Chain received the Nobel Prize in 1945 for the discovery and production of penicillin.

The discovery of penicillin stimulated the search for other antibiotics. Selman Waksman announced in 1944 that he had found a new antibiotic, streptomycin, produced by the actinomycete *Streptomyces griseus*. This discovery arose from the patient screening of about 10,000 strains of soil bacteria and fungi. Waksman received the Nobel Prize in 1952, and his success led to a worldwide search for other antibiotic-producing soil microorganisms. Microorganisms producing chloramphenicol, neomycin, terramycin, and tetracycline were isolated by 1953.

The discovery of chemotherapeutic agents and the development of newer, more powerful drugs has transformed modern medicine and greatly alleviated human suffering. Furthermore, antibiotics have proven exceptionally useful in microbiological research (Box 35.1).

1. What are chemotherapeutic agents? Antibiotics?
2. What contributions to chemotherapy were made by Ehrlich, Domagk, Fleming, Florey and Chain, and Waksman?

35.2 General Characteristics of Antimicrobial Drugs

As Ehrlich so clearly saw, to be successful a chemotherapeutic agent must have **selective toxicity**: it must kill or inhibit the microbial pathogen while damaging the host as little as possible. The degree of selective toxicity may be expressed in terms of (1) the therapeutic dose, the drug level required for clinical treatment of a particular infection, and (2) the toxic dose, the drug level at which the agent becomes too toxic for the host. The **therapeutic index** is the ratio of the toxic dose to the therapeutic dose. The larger the therapeutic index, the better the chemotherapeutic agent (all other things being equal).

A drug that disrupts a microbial function not found in eucaryotic animal cells often has a greater selective toxicity and a higher therapeutic index. For example, penicillin inhibits bacterial cell wall peptidoglycan synthesis but has little effect on host cells because they lack cell walls; therefore penicillin's therapeutic index

Table 35.1 Properties of Some Common Antibacterial Drugs

Drug	Primary Effect	Spectrum	Side Effects ^a
Ampicillin	Cidal	Broad (gram +, some -)	Allergic responses (diarrhea, anemia)
Bacitracin	Cidal	Narrow (gram +)	Renal injury if injected
Carbenicillin	Cidal	Broad (gram +, many -)	Allergic responses (nausea, anemia)
Cephalosporins	Cidal	Broad (gram +, some -)	(Allergic responses, thrombophlebitis, renal injury)
Chloramphenicol	Static	Broad (gram +, -; rickettsia and chlamydia)	Depressed bone marrow function, allergic reactions
Ciprofloxacin	Cidal	Broad (gram +, -)	Gastrointestinal upset, allergic responses
Clindamycin	Static	Narrow (gram +, anaerobes)	Diarrhea
Dapsone	Static	Narrow (mycobacteria)	(Anemia, allergic responses)
Erythromycin	Static	Narrow (gram +, mycoplasma)	(Gastrointestinal upset, hepatic injury)
Gentamicin	Cidal	Narrow (gram -)	(Allergic responses, nausea, loss of hearing, renal damage)
Isoniazid	Static or cidal	Narrow (mycobacteria)	(Allergic reactions, gastrointestinal upset, hepatic injury)
Methicillin	Cidal	Narrow (gram +)	Allergic responses (renal toxicity, anemia)
Penicillin	Cidal	Narrow (gram +)	Allergic responses (nausea, anemia)
Polymyxin B	Cidal	Narrow (gram -)	(Renal damage, neurotoxic reactions)
Rifampin	Static	Broad (gram +, mycobacteria)	(Hepatic injury, nausea, allergic responses)
Streptomycin	Cidal	Broad (gram +, -; mycobacteria)	(Allergic responses, nausea, loss of hearing, renal damage)
Sulfonamides	Static	Broad (gram +, -)	Allergic responses (renal and hepatic injury, anemia)
Tetracyclines	Static	Broad (gram +, -; rickettsia and chlamydia)	Gastrointestinal upset, teeth discoloration (renal and hepatic injury)
Trimethoprim	Cidal	Broad (gram +, -)	(Allergic responses, rash, nausea, leukopenia)
Vancomycin	Cidal	Narrow (gram +)	Hypotension, neutropenia, kidney damage, allergic reactions

^aOccasional side effects are in parentheses. Other side effects not listed may also arise.

is high. A drug may have a low therapeutic index because it inhibits the same process in host cells or damages the host in other ways. These undesirable effects on the host, called **side effects**, are of many kinds and may involve almost any organ system (**table 35.1**). Because side effects can be severe, chemotherapeutic agents should be administered with great care.

Drugs vary considerably in their range of effectiveness. Many are **narrow-spectrum drugs**—that is, they are effective only against a limited variety of pathogens (**table 35.1**). Others are **broad-spectrum drugs** and attack many different kinds of pathogens. Drugs may also be classified based on the general microbial group they act against: antibacterial, antifungal, antiprotozoan, and antiviral. Some agents can be used against more than one group; for example, sulfonamides are active against bacteria and some protozoa.

Chemotherapeutic agents can be synthesized by microorganisms or manufactured by chemical procedures independent of microbial activity. A number of the most commonly employed antibiotics are natural—that is, totally synthesized by one of a few bacteria or fungi (**table 35.2**). In contrast, several important chemotherapeutic agents are completely synthetic. The synthetic antibacterial drugs in **table 35.1** are the sulfonamides, trimethoprim, chloramphenicol, ciprofloxacin, isoniazid, and dapsone. Many antiviral and antiprotozoan drugs are synthetic. An increasing number of antibiotics are semisynthetic. Semisynthetic antibiotics are natural antibiotics that have been chemically modified by the addition of extra chemical groups to make them less susceptible to inactivation by pathogens. Ampicillin, carbenicillin, and methicillin (**table 35.1**) are good examples.

Table 35.2 Microbial Sources of Some Antibiotics

Microorganism	Antibiotic
Bacteria	
<i>Streptomyces</i> spp.	Amphotericin B Chloramphenicol (also synthetic) Erythromycin Kanamycin Neomycin Nystatin Rifampin Streptomycin Tetracyclines Vancomycin
<i>Micromonospora</i> spp.	Gentamicin
<i>Bacillus</i> spp.	Bacitracin Polymyxins
Fungi	
<i>Penicillium</i> spp.	Griseofulvin Penicillin
<i>Cephalosporium</i> spp.	Cephalosporins

Chemotherapeutic agents, like disinfectants, can be either **cidal** or **static**. As mentioned earlier (*see section 7.1*), static agents reversibly inhibit growth; if the agent is removed, the microorganisms will recover and grow again.

Although a cidal agent kills the target pathogen, its activity is concentration dependent and the agent may be only static at low

levels. The effect of an agent also varies with the target species: an agent may be cidal for one species and static for another. Because static agents do not directly destroy the pathogen, elimination of the infection depends on the host's own resistance mechanisms. A static agent may not be effective if the host's resistance is too low.

Host resistance and the immune response (chapters 31–33)

Some idea of the effectiveness of a chemotherapeutic agent against a pathogen can be obtained from the **minimal inhibitory concentration (MIC)**. The MIC is the lowest concentration of a drug that prevents growth of a particular pathogen. The **minimal lethal concentration (MLC)** is the lowest drug concentration that kills the pathogen. A cidal drug kills pathogens at levels only two to four times the MIC, whereas a static agent kills at much higher concentrations (if at all).

1. Define the following terms: selective toxicity, therapeutic index, side effect, narrow-spectrum drug, broad-spectrum drug, synthetic and semisynthetic antibiotics, cidal and static agents, minimal inhibitory concentration (MIC), and minimal lethal concentration (MLC).

35.3 Determining the Level of Antimicrobial Activity

Determination of antimicrobial effectiveness against specific pathogens is essential to proper therapy. Testing can show which agents are most effective against a pathogen and give an estimate of the proper therapeutic dose.

Dilution Susceptibility Tests

Dilution susceptibility tests can be used to determine MIC and MLC values. In the broth dilution test, a series of broth tubes (usually Mueller-Hinton broth) containing antibiotic concentrations in the range of 0.1 to 128 $\mu\text{g/ml}$ is prepared and inoculated with standard numbers of the test organism. The lowest concentration of the antibiotic resulting in no growth after 16 to 20 hours of incubation is the MIC. The MLC can be ascertained if the tubes showing no growth are subcultured into fresh medium lacking antibiotic. The lowest antibiotic concentration from which the microorganisms do not recover and grow when transferred to fresh medium is the MLC. The agar dilution test is very similar to the broth dilution test. Plates containing Mueller-Hinton agar and various amounts of antibiotic are inoculated and examined for growth. Recently several automated systems for susceptibility testing and MIC determination with broth or agar cultures have been developed.

Disk Diffusion Tests

If a rapidly growing aerobic or facultative pathogen like *Staphylococcus* or *Pseudomonas* is being tested, a disk diffusion technique may be used to save time and media. The principle behind the assay technique is fairly simple. When an antibiotic-impregnated disk is placed on agar previously inoculated with the test bacterium, the disk picks up moisture and the antibiotic diffuses radi-

ally outward through the agar, producing an antibiotic concentration gradient. The antibiotic is present at high concentrations near the disk and affects even minimally susceptible microorganisms (resistant organisms will grow up to the disk). As the distance from the disk increases, the antibiotic concentration drops and only more susceptible pathogens are harmed. A clear zone or ring is present around an antibiotic disk after incubation if the agent inhibits bacterial growth. The wider the zone surrounding a disk, the more susceptible the pathogen is. Zone width also is a function of the antibiotic's initial concentration, its solubility, and its diffusion rate through agar. Thus zone width cannot be used to compare directly the effectiveness of two different antibiotics.

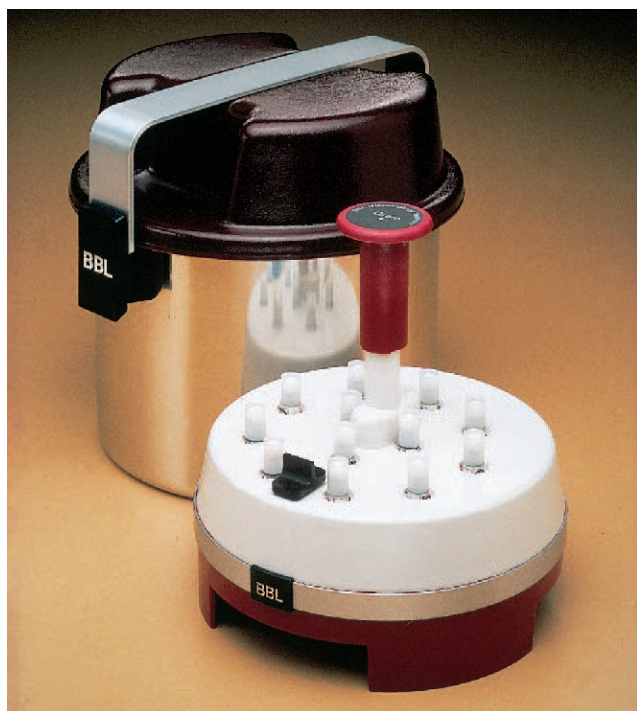
Currently the disk diffusion test most often used is the **Kirby-Bauer method**, which was developed in the early 1960s at the University of Washington Medical School by William Kirby, A. W. Bauer, and their colleagues. An inoculating loop or needle is touched to four or five isolated colonies of the pathogen growing on agar and then used to inoculate a tube of culture broth. The culture is incubated for a few hours at 35°C until it becomes slightly turbid and is diluted to match a turbidity standard. A sterile cotton swab is dipped into the standardized bacterial test suspension and used to evenly inoculate the entire surface of a Mueller-Hinton agar plate. After the agar surface has dried for about 5 minutes, the appropriate antibiotic test disks are placed on it, either with sterilized forceps or with a multiple applicator device (**figure 35.1**). The plate is immediately placed in a 35°C incubator. After 16 to 18 hours of incubation, the diameters of the zones of inhibition are measured to the nearest mm.

Kirby-Bauer test results are interpreted using a table that relates zone diameter to the degree of microbial resistance (table 35.3). The values in **table 35.3** were derived by finding the MIC values and zone diameters for many different microbial strains. A plot of MIC (on a logarithmic scale) versus zone inhibition diameter (arithmetic scale) is prepared for each antibiotic (**figure 35.2**). These plots are then used to find the zone diameters corresponding to the drug concentrations actually reached in the body. If the zone diameter for the lowest level reached in the body is smaller than that seen with the test pathogen, the pathogen should have an MIC value low enough to be destroyed by the drug. A pathogen with too high an MIC value (too small a zone diameter) is resistant to the agent at normal body concentrations.

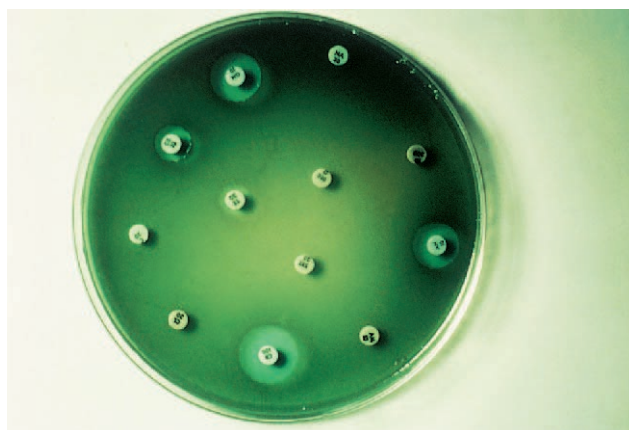
Measurement of Drug Concentrations in the Blood

A drug must reach a concentration at the site of infection above the pathogen's MIC to be effective. In cases of severe, life-threatening disease, it often is necessary to monitor the concentration of drugs in the blood and other body fluids. This may be achieved by microbiological, chemical, immunologic, enzymatic, or chromatographic assays.

1. How can dilution susceptibility tests and disk diffusion tests be used to determine microbial drug sensitivity?
2. Briefly describe the Kirby-Bauer test and its purpose.



(a)



(b)

Figure 35.1 The Kirby-Bauer Method. (a) A multiple antibiotic disk dispenser and (b) disk diffusion test results.

Table 35.3 Inhibition Zone Diameter of Selected Chemotherapeutic Drugs

Chemotherapeutic Drug	Disk Content	Zone Diameter (Nearest mm)		
		Resistant	Intermediate	Susceptible
Carbenicillin (with <i>Proteus</i> spp. and <i>E. coli</i>)	100 µg	≤17	18–22	≥23
Carbenicillin (with <i>Pseudomonas aeruginosa</i>)	100 µg	≤13	14–16	≥17
Ceftriaxone	30 µg	≤13	14–20	≥21
Chloramphenicol	30 µg	≤12	13–17	≥18
Erythromycin	15 µg	≤13	14–17	≥18
Penicillin G (with staphylococci)	10 U ^a	≤20	21–28	≥29
Penicillin G (with other microorganisms)	10 U	≤11	12–21	≥22
Streptomycin	10 µg	≤11	12–14	≥15
Sulfonamides	250 or 300 µg	≤12	13–16	≥17
Tetracycline	30 µg	≤14	15–18	≥19

^aOne milligram of penicillin G sodium = 1,600 units.

35.4 Mechanisms of Action of Antimicrobial Agents

The mechanisms of action of specific chemotherapeutic agents are taken up in more detail when individual drugs and groups of drugs are discussed later in this chapter. A few general observations are offered at this point. It is important to know something about the mechanisms of drug action because such knowledge helps explain the nature and degree of selective toxicity of individual drugs and sometimes aids in the design of new chemotherapeutic agents.

Antimicrobial drugs can damage pathogens in several ways, as can be seen in **table 35.4**, which summarizes the mechanisms of the antibacterial drugs listed in table 35.1. The most selective antibiotics are those that interfere with the synthesis of bacterial cell walls (e.g., penicillins, cephalosporins, vancomycin, and bacitracin). These drugs have a high therapeutic index because bacterial cell walls have a unique structure (see section 3.5) not found in eucaryotic cells.

Streptomycin, gentamicin, spectinomycin, clindamycin, chloramphenicol, tetracyclines, erythromycin, and many other

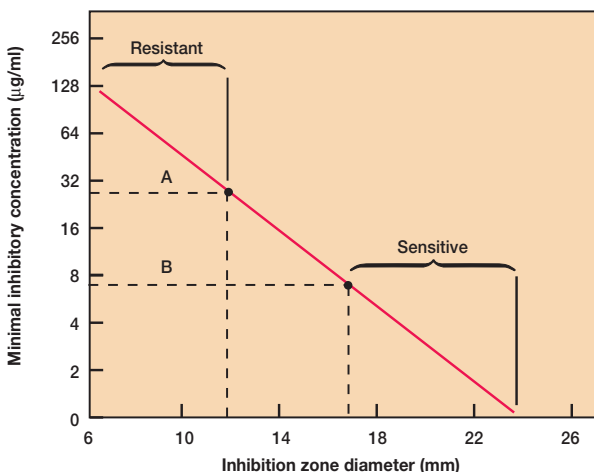


Figure 35.2 Interpretation of Kirby-Bauer Test Results. The relationship between the minimal inhibitory concentrations with a hypothetical drug and the size of the zone around a disk in which microbial growth is inhibited. As the sensitivity of microorganisms to the drug increases, the MIC value decreases and the inhibition zone grows larger. Suppose that this drug varies from 7–28 $\mu\text{g/ml}$ in the body during treatment. Dashed line A shows that any pathogen with a zone of inhibition less than 12 mm in diameter will have an MIC value greater than 28 $\mu\text{g/ml}$ and will be resistant to drug treatment. A pathogen with a zone diameter greater than 17 mm will have an MIC less than 7 $\mu\text{g/ml}$ and will be sensitive to the drug (see line B). Zone diameters between 12 and 17 mm indicate intermediate sensitivity and usually signify resistance.

Table 35.4 Mechanisms of Antibacterial Drug Action

Drug	Mechanism of Action
Cell Wall Synthesis Inhibition	
Penicillin	Inhibit transpeptidation enzymes involved in the cross-linking of the polysaccharide chains of the bacterial cell wall peptidoglycan. Activate cell wall lytic enzymes.
Ampicillin	
Carbenicillin	
Methicillin	
Cephalosporins	
Vancomycin	Binds directly to the D-Ala-D-Ala terminus and inhibits transpeptidation.
Bacitracin	Inhibits cell wall synthesis by interfering with action of the lipid carrier that transports wall precursors across the plasma membrane.
Protein Synthesis Inhibition	
Streptomycin	Binds with the 30S subunit of the bacterial ribosome to inhibit protein synthesis and causes misreading of mRNA.
Gentamicin	
Chloramphenicol	Binds to the 50S ribosomal subunit and blocks peptide bond formation through inhibition of peptidyl transferase.
Tetracyclines	Bind to the 30S ribosomal subunit and interfere with aminoacyl-tRNA binding.
Erythromycin and clindamycin	Bind to the 50S ribosomal subunit and inhibit peptide chain elongation.
Fusidic acid	Binds to EF-G and blocks translocation.
Nucleic Acid Synthesis Inhibition	
Ciprofloxacin and other quinolones	Inhibit bacterial DNA gyrase and thus interfere with DNA replication, transcription, and other activities involving DNA.
Rifampin	Blocks RNA synthesis by binding to and inhibiting the DNA-dependent RNA polymerase.
Cell Membrane Disruption	
Polymyxin B	Binds to the plasma membrane and disrupts its structure and permeability properties.
Metabolic Antagonism	
Sulfonamides	Inhibit folic acid synthesis by competition with <i>p</i> -aminobenzoic acid.
Trimethoprim	Blocks tetrahydrofolate synthesis through inhibition of the enzyme dihydrofolate reductase.
Dapsone	Interferes with folic acid synthesis.
Isoniazid	May disrupt pyridoxal or NAD metabolism and functioning. Inhibits the synthesis of the mycolic acid “cord factor.”

antibiotics inhibit protein synthesis by binding with the procaryotic ribosome. Because these drugs discriminate between procaryotic and eucaryotic ribosomes, their therapeutic index is fairly high, but not as favorable as that of cell wall synthesis inhibitors. Some drugs bind to the 30S (small) subunit, while others attach to the 50S (large) ribosomal subunit. Several different steps in the protein synthesis mechanism can be affected: aminoacyl-tRNA

binding, peptide bond formation, mRNA reading, and translocation. For example, fusidic acid binds to EF-G and blocks translocation, whereas mucopirocin inhibits isoleucyl-tRNA synthetase (see section 12.2).

The antibacterial drugs that inhibit nucleic acid synthesis or damage cell membranes often are not as selectively toxic as other antibiotics. This is because procaryotes and eucaryotes do not differ

as greatly with respect to nucleic acid synthetic mechanisms or cell membrane structure. Good examples of drugs that affect nucleic acid synthesis or membrane structure are quinolones and polymyxins. Quinolones inhibit the DNA gyrase and thus interfere with DNA replication, repair, and transcription. Polymyxins act as detergents or surfactants and disrupt the bacterial plasma membrane.

Several valuable drugs act as **antimetabolites**: they block the functioning of metabolic pathways by competitively inhibiting the use of metabolites by key enzymes. Sulfonamides and several other drugs inhibit folic acid metabolism. Sulfonamides (e.g., sulfanilamide, sulfamethoxazole, and sulfacetamide) have a high therapeutic index because humans cannot synthesize folic acid and must obtain it in their diet. Most bacterial pathogens synthesize their own folic acid and are therefore susceptible to inhibitors of folate metabolism. Antimetabolite drugs also can inhibit other pathways. For example, isoniazid interferes with either pyridoxal or NAD metabolism.

1. Give five ways in which chemotherapeutic agents kill or damage bacterial pathogens.
2. What are antimetabolites?

35.5 Factors Influencing the Effectiveness of Antimicrobial Drugs

It is crucial to recognize that drug therapy is not a simple matter. Drugs may be administered in several different ways, and they do not always spread rapidly throughout the body or immediately kill all invading pathogens. A complex array of factors influence the effectiveness of drugs.

First, the drug must actually be able to reach the site of infection. The mode of administration plays an important role. A drug such as penicillin G is not suitable for oral administration because it is relatively unstable in stomach acid. Some antibiotics—for example, gentamicin and other aminoglycosides—are not well absorbed from the intestinal tract and must be injected intramuscularly or given intravenously. Other antibiotics (neomycin, bacitracin) are applied topically to skin lesions. Nonoral routes of administration often are called **parenteral routes**. Even when an agent is administered properly, it may be excluded from the site of infection. For example, blood clots or necrotic tissue can protect bacteria from a drug, either because body fluids containing the agent may not easily reach the pathogens or because the agent is absorbed by materials surrounding it.

Second, the pathogen must be susceptible to the drug. Bacteria in abscesses may be dormant and therefore resistant to chemotherapy, because penicillins and many other agents affect pathogens only if they are actively growing and dividing. A pathogen, even though growing, may simply not be susceptible to a particular agent. For example, penicillins and cephalosporins, which inhibit cell wall synthesis (table 35.4), do not harm mycoplasmas, which lack cell walls.

Third, the chemotherapeutic agent must exceed the pathogen's MIC value if it is going to be effective. The concentration reached will depend on the amount of drug administered, the route of administration and speed of uptake, and the rate at which the drug is cleared or eliminated from the body. It makes sense that a drug will remain at high concentrations longer if it is absorbed over a long period and excreted slowly.

Finally, chemotherapy has been rendered a less effective and much more complex matter by the spread of drug resistance plasmids. The nature of drug resistance is dealt with later in this chapter.

1. Briefly discuss the factors that influence the effectiveness of antimicrobial drugs.
2. What is parenteral administration of a drug?

35.6 Antibacterial Drugs

With the background provided by the preceding introduction to chemotherapy, a few of the more important chemotherapeutic agents can be briefly described. Tables 35.1 and 35.4 summarize the properties and mechanism of action of a variety of drugs.

Sulfonamides or Sulfa Drugs

A good way to inhibit or kill pathogens is by use of compounds that are **structural analogues**, molecules structurally similar to metabolic intermediates. These analogues compete with metabolites in metabolic processes because of their similarity, but are just different enough so that they cannot function normally in cellular metabolism. The first antimetabolites to be used successfully as chemotherapeutic agents were the sulfonamides, discovered by G. Domagk. **Sulfonamides** or sulfa drugs are structurally related to sulfanilamide, an analogue of *p*-aminobenzoic acid (**figures 35.3 and 35.4**). The latter substance is used in the synthesis of the cofactor folic acid.

When sulfanilamide or another sulfonamide enters a bacterial cell, it competes with *p*-aminobenzoic acid for the active site of an enzyme involved in folic acid synthesis, and the folate concentration decreases. The decline in folic acid is detrimental to the bacterium because folic acid is essential to the synthesis of purines and pyrimidines, the bases used in the construction of DNA, RNA, and other important cell constituents. The resulting inhibition of purine and pyrimidine synthesis leads to cessation of bacterial growth or death of the pathogen. Sulfonamides are selectively toxic for many pathogens because these bacteria manufacture their own folate and cannot effectively take up the cofactor. In contrast, humans cannot synthesize folate and must obtain it in the diet; therefore sulfonamides will not affect the host. [Sulfonamides and the competitive inhibition of enzymes \(p. 164\); Purine and pyrimidine structure and synthesis \(pp. 216–18\)](#)

The effectiveness of sulfonamides is limited by the increasing sulfonamide resistance of many bacteria. Furthermore, as many as 5% of the patients receiving sulfa drugs experience adverse side effects, chiefly in the form of allergic responses (fever, hives, and rashes).

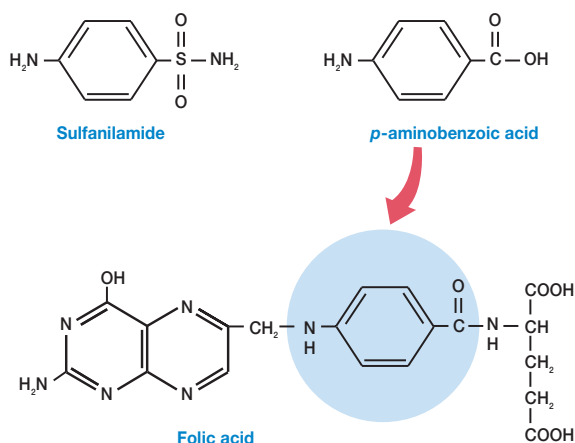


Figure 35.3 Sulfanilamide. Sulfanilamide and its relationship to the structure of folic acid.

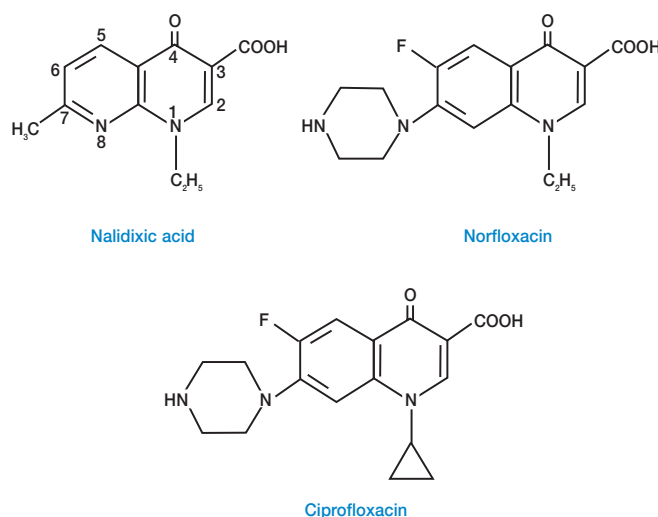


Figure 35.5 Quinolone Antimicrobial Agents. Ciprofloxacin and norfloxacin are newer generation fluoroquinolones. The 4-quinolone ring in nalidixic acid has been numbered.

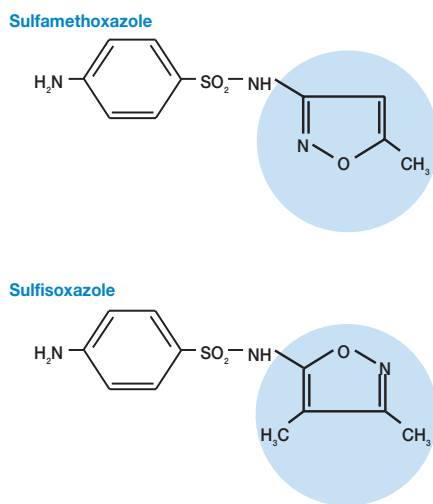


Figure 35.4 Two Modern Sulfonamide Drugs. The shaded areas are side chains substituted for a hydrogen in sulfanilamide (figure 35.3).

Quinolones

A second group of synthetic antimicrobial agents are increasingly used to treat a wide variety of infections. The **quinolones** are synthetic drugs that contain the 4-quinolone ring. The first quinolone, nalidixic acid (**figure 35.5**), was synthesized in 1962. More recently a family of fluoroquinolones has been produced. Three of these—ciprofloxacin, norfloxacin, and ofloxacin—are currently

used in the United States, and more fluoroquinolones are being synthesized and tested. Quinolones are effective when administered orally. They sometimes cause adverse side effects, particularly gastrointestinal upset.

Quinolones act by inhibiting the bacterial DNA gyrase or topoisomerase II, probably by binding to the DNA gyrase complex. This enzyme introduces negative twists in DNA and helps separate its strands (**figure 35.6**). DNA gyrase inhibition disrupts DNA replication and repair, transcription, bacterial chromosome separation during division, and other cell processes involving DNA. Fluoroquinolones also inhibit topoisomerase IV, another enzyme that untangles DNA during replication. It is not surprising that quinolones are bactericidal. [The mechanism of DNA replication \(pp. 236–39\)](#)

The quinolones are broad-spectrum drugs. They are highly effective against enteric bacteria such as *E. coli* and *Klebsiella pneumoniae*. They can be used with *Haemophilus*, *Neisseria*, *Pseudomonas aeruginosa*, and other gram-negative pathogens. The quinolones also are active against gram-positive bacteria such as *Staphylococcus aureus*, *Streptococcus pyogenes*, and *Mycobacterium tuberculosis*. Currently they are used in treating urinary tract infections, sexually transmitted diseases caused by *Neisseria* and *Chlamydia*, gastrointestinal infections, respiratory tract infections, skin infections, and osteomyelitis.

Penicillins

Penicillin G or benzylpenicillin, the first antibiotic to be widely used in medicine, has the structural properties characteristic of

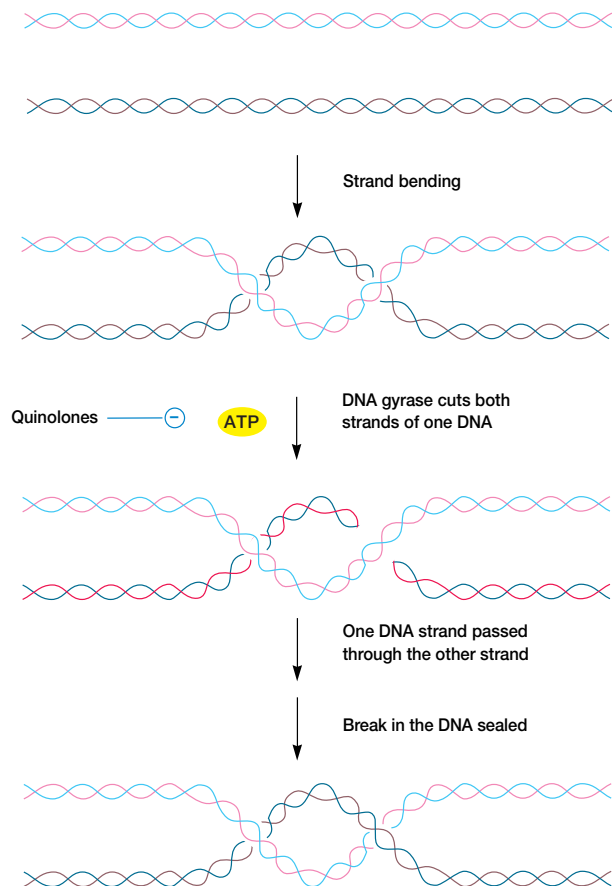


Figure 35.6 DNA Gyrase Action and Quinolone Inhibition.

the penicillin family (figure 35.7). Most **penicillins** are derivatives of 6-aminopenicillanic acid and differ from one another only with respect to the side chain attached to its amino group. The most crucial feature of the molecule is the β -lactam ring, which appears to be essential for activity. **Penicillinase**, the enzyme synthesized by many penicillin-resistant bacteria, destroys penicillin activity by hydrolyzing a bond in this ring (figure 35.7).

The mechanism of action of penicillins is still not completely known. Their structures do resemble that of the terminal D-alanyl-D-alanine found on the peptide side chain of the peptidoglycan subunit. It has been proposed that penicillins inhibit the enzyme catalyzing the transpeptidation reaction because of their structural similarity, which would block the synthesis of a complete, fully cross-linked peptidoglycan and lead to osmotic lysis. The mechanism is consistent with the observation that penicillins act only on growing bacteria that are synthesizing new peptidoglycan. However, more recently it has been discovered that penicillins bind to several penicillin-binding proteins

and may destroy bacteria by activating their own autolytic enzymes. There now is some evidence that penicillin kills bacteria even in the absence of autolysins or murein hydrolases. Lysis could occur after bacterial viability has already been lost. Penicillin may stimulate special proteins called bacterial holins to form holes or lesions in the plasma membrane. This would directly lead to membrane leakage and death; murein hydrolases also could move through the holes, disrupt the peptidoglycan, and lyse the cell. Apparently the mechanism of penicillin action is more complex than previously imagined. [Bacterial peptidoglycan structure \(pp. 56–57\); Peptidoglycan synthesis \(pp. 221–22\)](#)

Penicillins differ from each other in several ways. Penicillin G is effective against gonococci, meningococci, and several gram-positive pathogens such as streptococci and staphylococci ([table 35.5](#)), but it must be administered parenterally because it is destroyed by stomach acid. Penicillin V is similar to penicillin G, but it is more acid resistant and can be given orally. Ampicillin can be administered orally and has a broader spectrum of activity as it is effective against gram-negative bacteria such as *Haemophilus*, *Salmonella*, and *Shigella*. Carbenicillin and ticarcillin also are broad spectrum and particularly potent against *Pseudomonas* and *Proteus*.

An increasing number of bacteria are penicillin resistant. Penicillinase-resistant penicillins such as methicillin (figure 35.7), nafcillin, and oxacillin are frequently employed against these bacterial pathogens.

Although penicillins are the least toxic of the antibiotics, about 1 to 5% of the adults in the United States are allergic to them. Occasionally a person will die of a violent allergic response; therefore patients should be questioned about penicillin allergies before treatment is begun.

Cephalosporins

Cephalosporins are a family of antibiotics originally isolated in 1948 from the fungus *Cephalosporium*, and their β -lactam structure is very similar to that of the penicillins ([figure 35.8](#)). As might be expected from their structural similarities, cephalosporins resemble penicillins in inhibiting the transpeptidation reaction during peptidoglycan synthesis. They are broad-spectrum drugs frequently given to patients with penicillin allergies.

Many cephalosporins are in use. There are three groups or generations of these drugs that differ in their spectrum of activity (figure 35.8). First-generation cephalosporins are more effective against gram-positive than gram-negative pathogens. Second-generation drugs act against many gram-negative as well as gram-positive pathogens. Third-generation drugs are particularly effective against gram-negative pathogens, and often also reach the central nervous system.

Most cephalosporins (including cephalothin, cefoxitin, ceftriaxone, and cefoperazone) are administered parenterally. Cefoperazone is resistant to destruction by β -lactamases and effective against many gram-negative bacteria, including *Pseudomonas aeruginosa*. Cephalexine and cefixime are given orally rather than by injection.

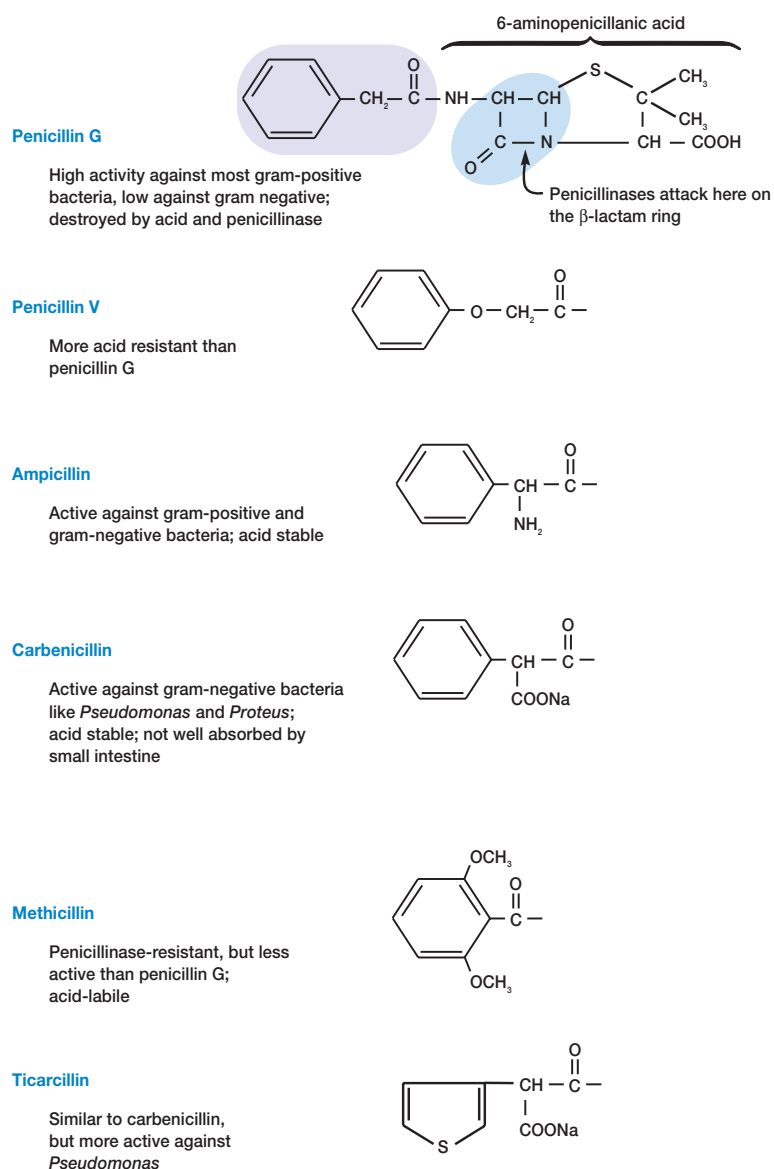


Figure 35.7 Penicillins. The structures and characteristics of representative penicillins. All are derivatives of 6-aminopenicillanic acid; in each case the purple shaded portion of penicillin G is replaced by the side chain indicated. The β -lactam ring is also shaded (blue), and an arrow points to the bond that is hydrolyzed by penicillinase.

The Tetracyclines

The **tetracyclines** are a family of antibiotics with a common four-ring structure to which a variety of side chains are attached (**figure 35.9**). Oxytetracycline and chlortetracycline are naturally produced by some species of the actinomycete genus *Streptomyces*; others are semisynthetic drugs. These antibiotics in-

hibit protein synthesis by combining with the small (30S) subunit of the ribosome and inhibiting the binding of aminoacyl-tRNA molecules to the ribosomal A site. Because their action is only bacteriostatic, the effectiveness of treatment depends on active host resistance to the pathogen. [The mechanism of protein synthesis \(pp. 265–72\)](#)

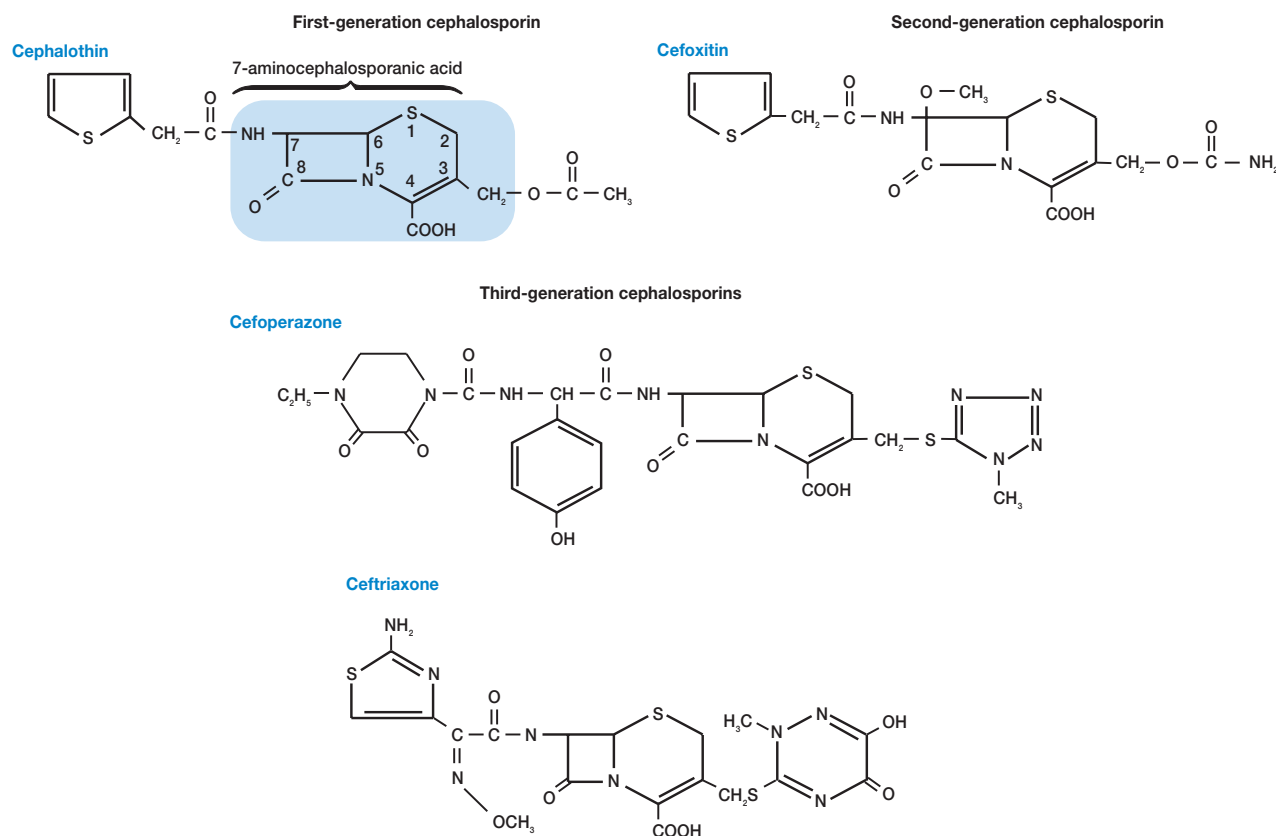


Figure 35.8 Cephalosporin Antibiotics. These drugs are derivatives of 7-aminocephalosporanic acid and contain a β -lactam ring.

Tetracycline (chlortetracycline, doxycycline)

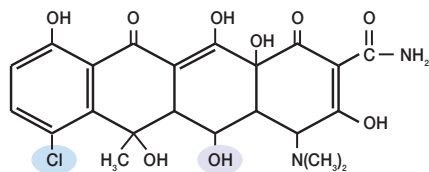


Figure 35.9 Tetracyclines. Three members of the tetracycline family. Tetracycline lacks both of the groups that are shaded. Chlortetracycline (aureomycin) differs from tetracycline in having a chlorine atom (blue); doxycycline consists of tetracycline with an extra hydroxyl (purple).

Tetracyclines are broad-spectrum antibiotics active against gram-negative bacteria, gram-positive bacteria, rickettsias, chlamydiae, and mycoplasmas. High doses may result in nausea, diarrhea, yellowing of teeth in children, and damage to the liver and kidneys.

Aminoglycoside Antibiotics

There are several important **aminoglycoside antibiotics**. **Streptomycin**, kanamycin, neomycin, and tobramycin are synthesized by *Streptomyces*, whereas gentamicin comes from a related bacterium, *Micromonospora purpurea*. Although there is considerable variation in structure among the different aminoglycosides, all contain a cyclohexane ring and amino sugars (**figure 35.10**). Aminoglycosides bind to the small ribosomal subunit and interfere with protein synthesis in at least two ways. They directly inhibit protein synthesis and also cause misreading of the genetic message carried by mRNA.

The aminoglycosides are bactericidal and tend to be most active against gram-negative pathogens. Streptomycin's usefulness has decreased greatly due to widespread drug resistance, but it is still effective against tuberculosis and plague. Gentamicin is used to treat *Proteus*, *Escherichia*, *Klebsiella*, and *Serratia* infections. Aminoglycosides are quite toxic and can cause deafness, renal damage, loss of balance, nausea, and allergic responses.

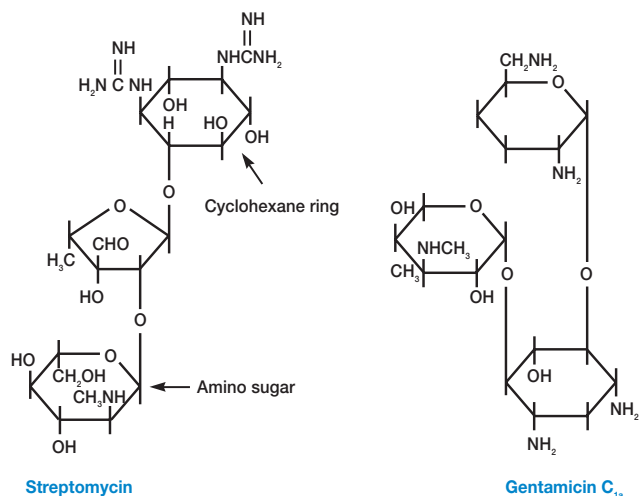


Figure 35.10 Representative Aminoglycoside Antibiotics.

Erythromycin and Other Macrolides

Erythromycin, the most frequently used **macrolide antibiotic**, is synthesized by *Streptomyces erythraeus*. The macrolides contain a 12- to 22-carbon lactone ring linked to one or more sugars (**figure 35.11**). Erythromycin is usually bacteriostatic and binds with the 23S rRNA of the 50S ribosomal subunit to inhibit peptide chain elongation during protein synthesis.

Erythromycin is a relatively broad-spectrum antibiotic effective against gram-positive bacteria, mycoplasmas, and a few gram-negative bacteria. It is used with patients allergic to penicillins and in the treatment of whooping cough, diphtheria, diarrhea caused by *Campylobacter*, and pneumonia from *Legionella* or *Mycoplasma* infections.

Newer macrolides are now in use. Clindamycin is effective against a variety of bacteria including staphylococci and anaerobes such as *Bacteroides*. Azithromycin is particularly effective against *Chlamydia trachomatis*.

Vancomycin and Teicoplanin

Vancomycin is a glycopeptide antibiotic produced by *Streptomyces orientalis*. It is a cup-shaped molecule composed of a peptide linked to a disaccharide. The antibiotic blocks peptidoglycan synthesis by inhibiting the transpeptidation step that cross-links adjacent peptidoglycan strands. The resulting peptidoglycan is mechanically weak and the cells osmotically lyse. Vancomycin's peptide portion binds specifically to the D-alanine-D-alanine terminal sequence on the pentapeptide portion of peptidoglycan. This complex blocks transpeptidase action.

The antibiotic is bactericidal for *Staphylococcus* and some members of the genera *Clostridium*, *Bacillus*, *Streptococcus*, and *Enterococcus*. It is given both orally and intravenously, and has been particularly important in the treatment of antibiotic resistant

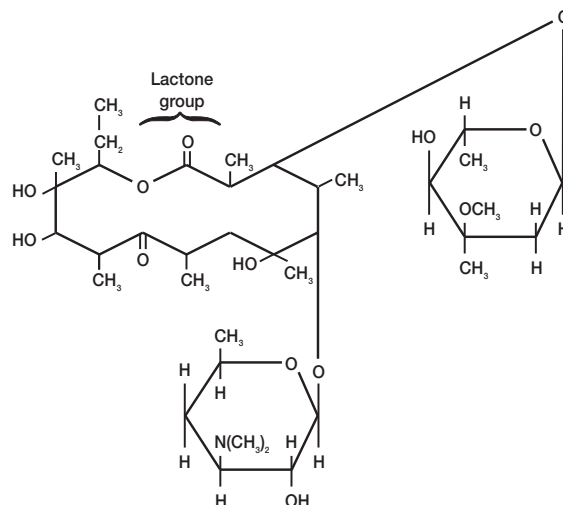


Figure 35.11 Erythromycin, a Macrolide Antibiotic. The 14-member lactone ring is connected to two sugars.

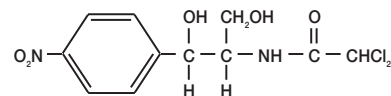


Figure 35.12 Chloramphenicol.

staphylococcal and enterococcal infections. Vancomycin-resistant strains of *Enterococcus* have become widespread and recently a few cases of resistant *Staphylococcus aureus* have appeared.

Teicoplanin is a glycopeptide antibiotic from *Actinoplanes teichomyceticus* that is similar in structure and mechanism of action to vancomycin. It is active against staphylococci, enterococci, streptococci, clostridia, *Listeria*, and many other gram-positive pathogens. This antibiotic presently is used in Europe and elsewhere, but not in the United States.

Chloramphenicol

Although **chloramphenicol** (**figure 35.12**) was first produced from cultures of *Streptomyces venezuelae*, it is now made through chemical synthesis. Like erythromycin, chloramphenicol binds to 23S rRNA on the 50S ribosomal subunit. It inhibits the peptidyl transferase and is bacteriostatic.

This antibiotic has a very broad spectrum of activity but unfortunately is quite toxic. One may see allergic responses or neurotoxic reactions. The most common side effect is a temporary or permanent depression of bone marrow function, leading to aplastic anemia and a decreased number of blood leukocytes. Chloramphenicol is used only in life-threatening situations when no other drug is adequate.

Box 35.2

Antibiotic Misuse and Drug Resistance

The sale of antimicrobial drugs is big business. In the United States millions of pounds of antibiotics valued at billions of dollars are produced annually. Approximately 40 to 50% of these antibiotics are added to livestock feed.

Because of the massive quantities of antibiotics being prepared and used, an increasing number of diseases are resisting treatment due to the spread of drug resistance. A good example is *Neisseria gonorrhoeae*, the causative agent of gonorrhea. Gonorrhea was first treated successfully with sulfonamides in 1936, but by 1942 most strains were resistant and physicians turned to penicillin. Within 16 years a penicillin-resistant strain had emerged in the Far East. A penicillinase-producing gonococcus reached the United States in 1976 and is still spreading in this country.

In late 1968 an epidemic of dysentery caused by *Shigella* broke out in Guatemala and affected at least 112,000 persons; 12,500 deaths resulted. The strains responsible for this devastation carried an R plasmid giving them resistance to chloramphenicol, tetracycline, streptomycin, and sulfonamide. In 1972 a typhoid epidemic swept through Mexico producing 100,000 infections and 14,000 deaths. It was due to a *Salmonella typhi* strain with the same multiple drug-resistance pattern seen in the previous *Shigella* outbreak.

Haemophilus influenzae type b is responsible for many cases of childhood pneumonia and middle ear infections, as well as respiratory infections and meningitis. It is now becoming increasingly resistant to tetracyclines, ampicillin, and chloramphenicol. The same situation is occurring with *Streptococcus pneumoniae*.

In 1946 almost all strains of *Staphylococcus* were penicillin sensitive. Today most hospital strains are resistant to penicillin G, and some are now also resistant to methicillin and/or gentamicin and only can be treated with vancomycin. Some strains of *Enterococcus* have become resistant to most antibiotics, including vancomycin. Recently a few cases of vancomycin-resistant *S. aureus* have been reported in the United States and Japan. At present these strains are only intermediately resistant to vancomycin. If full vancomycin resistance develops and spreads, *S. aureus* may become untreatable.

It is clear from these and other examples (e.g., multiresistant *Mycobacterium tuberculosis*) that drug resistance is an extremely serious public health problem. Much of the difficulty arises from drug misuse. Drugs frequently have been overused in the past. It has been estimated that over 50% of the antibiotic prescriptions in hospitals are given without clear evidence of infection or adequate medical indication. Many physicians have

administered antibacterial drugs to patients with colds, influenza, viral pneumonia, and other viral diseases. A recent study showed that over 50% of the patients diagnosed with colds and upper respiratory infections and 66% of those with chest colds (bronchitis) are given antibiotics, even though over 90% of these cases are caused by viruses. Frequently antibiotics are prescribed without culturing and identifying the pathogen or without determining bacterial sensitivity to the drug. Toxic, broad-spectrum antibiotics are sometimes given in place of narrow-spectrum drugs as a substitute for culture and sensitivity testing, with the consequent risk of dangerous side effects, superinfections, and the selection of drug-resistant mutants. The situation is made worse by patients not completing their course of medication. When antibiotic treatment is ended too early, drug-resistant mutants may survive. Drugs are available to the public in many countries; people may practice self-administration of antibiotics and further increase the prevalence of drug-resistant strains.

The use of antibiotics in animal feeds is undoubtedly another contributing factor to increasing drug resistance. The addition of low levels of antibiotics to livestock feeds does raise the efficiency and rate of weight gain in cattle, pigs, and chickens (partially because of infection control in overcrowded animal populations). However, this also increases the number of drug-resistant bacteria in animal intestinal tracts. There is evidence for the spread of bacteria such as *Salmonella* from animals to human populations. In 1983, 18 people in four midwestern states were infected with a multiple-drug-resistant strain of *Salmonella newport*. Eleven were hospitalized for salmonellosis and one died. All 18 patients had recently been infected by eating hamburger from beef cattle fed subtherapeutic doses of chlortetracycline for growth promotion. Resistance to some antibiotics has been traced to the use of specific farmyard antibiotics. Avoparcin resembles vancomycin in structure, and virginiamycin resembles Synercid. There is good circumstantial evidence that extensive use of these two antibiotics in animal feed has led to an increase in vancomycin and Synercid resistance among enterococci. The use of the quinolone antibiotic enrofloxacin in swine herds appears to have promoted ciprofloxacin resistance in pathogenic strains of *Salmonella*. Elimination of antibiotic food supplements might well aid in slowing the spread of drug resistance.

The spread of antibiotic resistance can be due to quite subtle factors. For example, products such as soap and deodorants often now contain triclosan and other germicides. There is increasing evidence that the widespread use of triclosan actually favors an increase in antibiotic resistance (see section 7.5).

35.7 Drug Resistance

The spread of drug-resistant pathogens is one of the most serious threats to the successful treatment of microbial disease (**Box 35.2**). This section describes the ways in which bacteria acquire drug resistance and how resistance spreads within a bacterial population.

Mechanisms of Drug Resistance

Bacteria become drug resistant in several different ways. It should be noted at the beginning that a particular type of resistance mechanism

1. For each class of antibacterial drugs presented, give the following information: general chemical composition or structure, mechanism of action, behavioral properties (cidal or static, broad spectrum or narrow spectrum), route of administration and therapeutic uses, significant problems with side effects, and drug resistance.
2. Define structural analogue, 6-aminopenicillanic acid, penicillinase, aminoglycoside, and macrolide.

is not confined to a single class of drugs. Two bacteria may use different resistance mechanisms to withstand the same chemotherapeutic agent. Furthermore, resistant mutants arise spontaneously and are then selected. Mutants are not created directly by exposure to a drug.

Pathogens often become resistant simply by preventing entrance of the drug. Many gram-negative bacteria are unaffected by penicillin G because it cannot penetrate the envelope's outer membrane. Changes in penicillin binding proteins also render a cell resistant. A decrease in permeability can lead to sulfonamide resistance. Mycobacteria resist many drugs because of the high content of mycolic acids (*see p. 543*) in a complex lipid layer outside their peptidoglycan. This layer is impermeable to most drugs.

A second resistance strategy is to pump the drug out of the cell after it has entered. Some pathogens have plasma membrane translocases, often called efflux pumps, that expel drugs. Because they are relatively nonspecific and can pump many different drugs, these transport proteins often are called multidrug-resistance pumps. Many are drug/proton antiporters—that is, protons enter the cell as the drug leaves. Such systems are present in *E. coli*, *Pseudomonas aeruginosa*, *Mycobacterium smegmatis*, and *Staphylococcus aureus*.

Many bacterial pathogens resist attack by inactivating drugs through chemical modification. The best-known example is the hydrolysis of the β -lactam ring of many penicillins by the enzyme penicillinase. Drugs also are inactivated by the addition of groups. Resistant organisms may phosphorylate or acetylate aminoglycosides and acetylate chloramphenicol.

Because each chemotherapeutic agent acts on a specific target, resistance arises when the target enzyme or organelle is modified so that it is no longer susceptible to the drug. For example, the affinity of ribosomes for erythromycin and chloramphenicol can be decreased by a change in the 23S rRNA to which they bind. Enterococci become resistant to vancomycin by changing the terminal D-alanine-D-alanine in their peptidoglycan to a D-alanine-D-lactate. This drastically reduces antibiotic binding. The effects of antimetabolites may be resisted through alteration of susceptible enzymes. In sulfonamide-resistant bacteria the enzyme that uses *p*-aminobenzoic acid during folic acid synthesis (the tetrahydropterotic acid synthetase) often has a much lower affinity for sulfonamides. *Mycobacterium tuberculosis* becomes resistant to the drug rifampin due to mutations that alter the β subunit of its RNA polymerase. Rifampin cannot bind to the mutant RNA polymerase and block the initiation of transcription.

Resistant bacteria may either use an alternate pathway to bypass the sequence inhibited by the agent or increase the production of the target metabolite. For example, some bacteria are resistant to sulfonamides simply because they use preformed folic acid from their surroundings rather than synthesize it themselves. Other strains increase their rate of folic acid production and thus counteract sulfonamide inhibition.

The Origin and Transmission of Drug Resistance

The genes for drug resistance are present on both the bacterial chromosome and **plasmids**, small DNA molecules that can exist separate from the chromosome or be integrated in it. [Plasmids \(pp. 294–97\)](#)

Spontaneous mutations in the bacterial chromosome, although they do not occur very often, will make bacteria drug resistant. Usually such mutations result in a change in the drug receptor; therefore the antibiotic cannot bind and inhibit (e.g., the streptomycin receptor protein on bacterial ribosomes). Many mutants are probably destroyed by natural host resistance mechanisms. However, when a patient is being treated extensively with antibiotics, some resistant mutants may survive and flourish because of their competitive advantage over nonresistant strains.

Frequently a bacterial pathogen is drug resistant because it has a plasmid bearing one or more resistance genes; such plasmids are called **R plasmids** (resistance plasmids). Plasmid resistance genes often code for enzymes that destroy or modify drugs; for example, the hydrolysis of penicillin or the acetylation of chloramphenicol and aminoglycoside drugs. Plasmid-associated genes have been implicated in resistance to the aminoglycosides, chloramphenicol, penicillins and cephalosporins, erythromycin, tetracyclines, sulfonamides, and others. Once a bacterial cell possesses an R plasmid, the plasmid may be transferred to other cells quite rapidly through normal gene exchange processes such as conjugation, transduction, and transformation. Because a single plasmid may carry genes for resistance to several drugs, a pathogen population can become resistant to several antibiotics simultaneously, even though the infected patient is being treated with only one drug. [Conjugation \(pp. 302–5\)](#); [Transduction \(pp. 307–9\)](#); [Transformation \(pp. 305–7\)](#)

Extensive drug treatment favors the development and spread of antibiotic-resistant strains because the antibiotic destroys normal, susceptible bacteria that would usually compete with drug-resistant strains. The result may be the emergence of drug-resistant pathogens leading to a **superinfection**. Superinfections are a significant problem because of the existence of multiple-drug-resistant bacteria that often produce drug-resistant respiratory and urinary tract infections. A classic example of a superinfection resulting from antibiotic administration is the disease pseudomembranous enterocolitis caused by *Clostridium difficile*. When a patient is treated with clindamycin, ampicillin, or cephalosporin, many intestinal bacteria are killed, but *C. difficile* is not. This intestinal inhabitant, which is normally a minor constituent of the population, flourishes in the absence of competition and produces a toxin that stimulates the secretion of a pseudomembrane by intestinal cells. If the superinfection is not treated early with vancomycin, the pseudomembrane must be surgically removed or the patient will die. Fungi, such as the yeast *Candida albicans*, also produce superinfections when bacterial competition is eliminated by antibiotics.

Several strategies can be employed to discourage the emergence of drug resistance. The drug can be given in a high enough concentration to destroy susceptible bacteria and most spontaneous mutants that might arise during treatment. Sometimes two different drugs can be administered simultaneously with the hope that each drug will prevent the emergence of resistance to the other. Finally, chemotherapeutic drugs, particularly broad-spectrum drugs, should be used only when definitely necessary. If possible, the pathogen should be identified, drug sensitivity tests run, and the proper narrow-spectrum drug employed.

Despite efforts to control the emergence and spread of drug resistance, the situation continues to worsen. Of course, as mentioned previously, antibiotics should be used in ways that reduce the development of resistance. Another approach is to search for new antibiotics that microorganisms have never encountered (*see Box 29.1*). Pharmaceutical companies collect and analyze samples from around the world in a search for completely new antimicrobial agents. Structure-based or rational drug design is a third option. If the three-dimensional structure of a susceptible target molecule such as an enzyme essential to microbial function is known, computer programs can be used to design drugs that precisely fit the target molecule. These drugs might be able to bind to the target and disrupt its function sufficiently to destroy the pathogen. Pharmaceutical companies are using this approach to attempt to develop drugs for the treatment of AIDS, cancer, and the common cold. At least one company is developing “enhancers.” These are cationic peptides that disrupt bacterial membranes by displacing their magnesium ions. Antibiotics then penetrate and rapidly exert their effects. Other pharmaceutical companies are developing efflux-pump inhibitors to administer with antibiotics and prevent their expulsion by the resistant pathogen.

There has been some recent progress in developing new antibiotics that are effective against drug-resistant pathogens. Two new drugs are fairly effective against vancomycin-resistant enterococci. Synercid is a mixture of the streptogramin antibiotics quinupristin and dalfopristin that inhibits protein synthesis. A second drug, linezolid (Zyvox), is the first drug in a new family of antibiotics, the oxazolidinones. It inhibits protein synthesis and is active against both vancomycin-resistant enterococci and methicillin-resistant *Staphylococcus aureus*.

A most interesting response to the current crisis is the renewed interest in an idea first proposed early in the twentieth century by Felix d’Herelle, one of the discoverers of bacterial viruses or bacteriophages (*see p. 363*). d’Herelle proposed that bacteriophages could be used to treat bacterial diseases. Although most microbiologists did not pursue his proposal actively due to technical difficulties and the advent of antibiotics, Russian scientists developed the medical use of bacteriophages. Currently Russian physicians use bacteriophages to treat many bacterial infections. Bandages are saturated with phage solutions, phage mixtures are administered orally, and phage preparations are given intravenously to treat *Staphylococcus* infections. Three American companies are actively conducting research on phage therapy and preparing to carry out clinical trials. [Bacteriophages \(chapter 17\)](#)

1. Briefly describe the five major ways in which bacteria become resistant to drugs and give an example of each.
2. Define plasmid, R plasmid, and superinfection. How are R plasmids involved in the spread of drug resistance?

35.8 Antifungal Drugs

Treatment of fungal infections generally has been less successful than that of bacterial infections largely because eucaryotic fungal cells are much more similar to human cells than are bacteria.

Many drugs that inhibit or kill fungi are therefore quite toxic for humans. In addition, most fungi have a detoxification system that modifies many antibiotics, probably by hydroxylation. As a result the added antibiotics are fungistatic only as long as repeated application maintains high levels of unmodified antibiotic. Despite their relatively low therapeutic index, a few drugs are useful in treating many major fungal diseases. Effective antifungal agents frequently either extract membrane sterols or prevent their synthesis. Similarly, because animal cells do not have cell walls, the enzyme chitin synthase is the target for fungal-active antibiotics such as polyoxin D and nikkomycin.

Fungal infections are often subdivided into infections of superficial tissues or superficial mycoses and systemic mycoses. Treatment for these two types of disease is very different. Several drugs are used to treat superficial mycoses. Three drugs containing imidazole—miconazole, ketoconazole (**figure 35.13**), and clotrimazole—are broad-spectrum agents available as creams and solutions for the treatment of dermatophyte infections such as athlete’s foot, and oral and vaginal candidiasis (*see chapter 40*). They are thought to disrupt fungal membrane permeability and inhibit sterol synthesis. Tolnaftate is used topically for the treatment of cutaneous infections, but is not as effective against infections of the skin and hair. **Nystatin** (*figure 35.13*), a polyene antibiotic from *Streptomyces*, is used to control *Candida* infections of the skin, vagina, or alimentary tract. **Griseofulvin** (*figure 35.13*), an antibiotic formed by *Penicillium*, is given orally to treat chronic dermatophyte infections. It is thought to disrupt the mitotic spindle and inhibit cell division; it also may inhibit protein and nucleic acid synthesis. Side effects of griseofulvin include headaches, gastrointestinal upset, and allergic reactions. [Superficial and systemic mycoses \(pp. 942–44, 945–48\)](#)

Systemic infections are very difficult to control and can be fatal. Three drugs commonly used against systemic mycoses are **amphotericin B**, 5-flucytosine, and fluconazole (*figure 35.13*). Amphotericin B from *Streptomyces* spp. binds to the sterols in fungal membranes, disrupting membrane permeability and causing leakage of cell constituents. It is quite toxic and used only for serious, life-threatening infections. The synthetic oral antimycotic agent 5-flucytosine (5-fluorocytosine) is effective against most systemic fungi, although drug resistance often develops rapidly. The drug is converted to 5-fluorouracil by the fungi, incorporated into RNA in place of uracil, and disrupts RNA function. Its side effects include skin rashes, diarrhea, nausea, aplastic anemia, and liver damage. Fluconazole is used in the treatment of candidiasis, cryptococcal meningitis, and coccidioid meningitis. Adverse effects are relatively uncommon.

Just as with antibacterial drugs, overuse of antifungal agents leads to an increase in drug resistance. For example, *Candida* infections are becoming more frequent and drug resistant.

1. Summarize the mechanism of action and the therapeutic use of the following antifungal drugs: miconazole, nystatin, griseofulvin, amphotericin B, and 5-flucytosine.

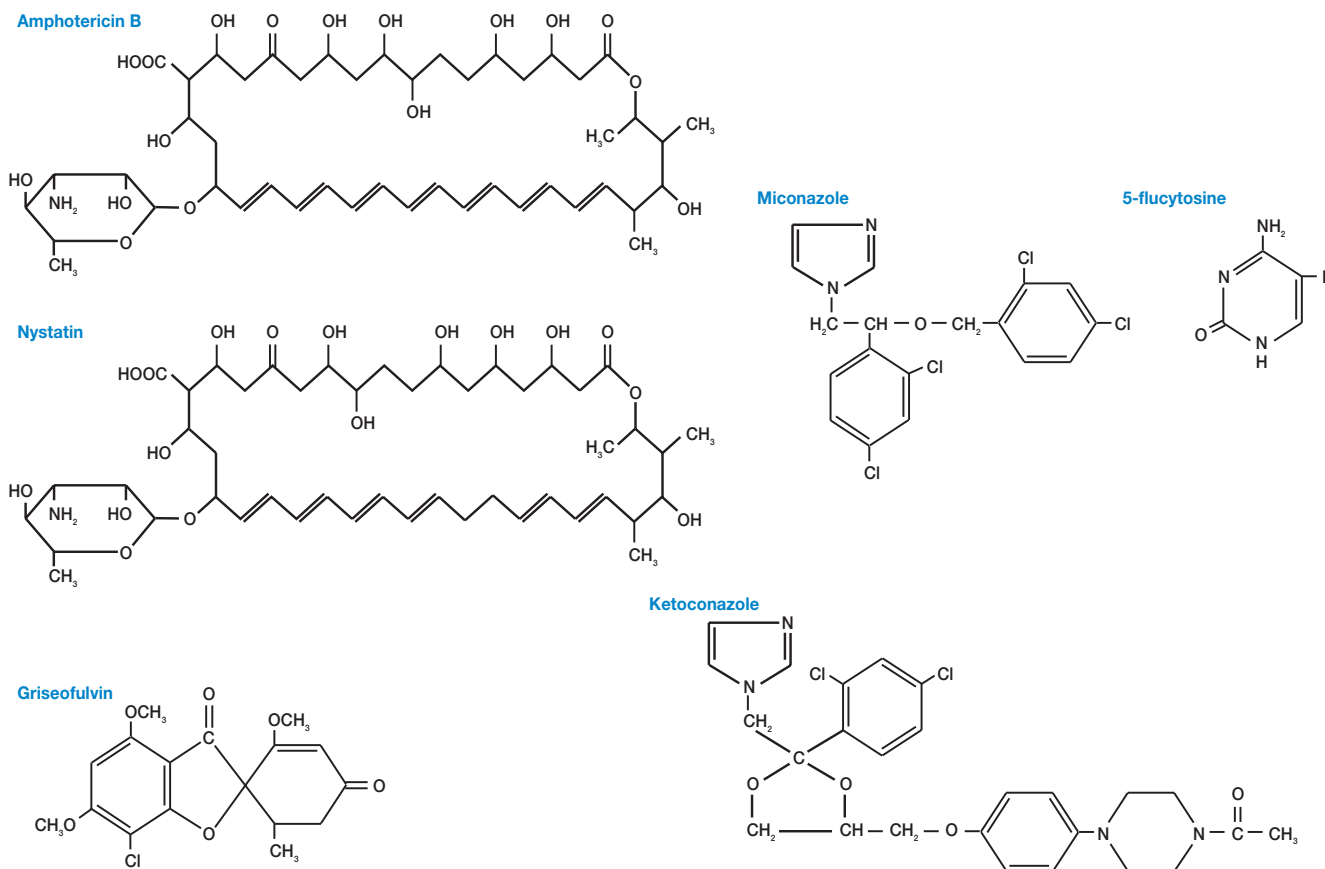


Figure 35.13 Antifungal Drugs. Six commonly used drugs are shown here.

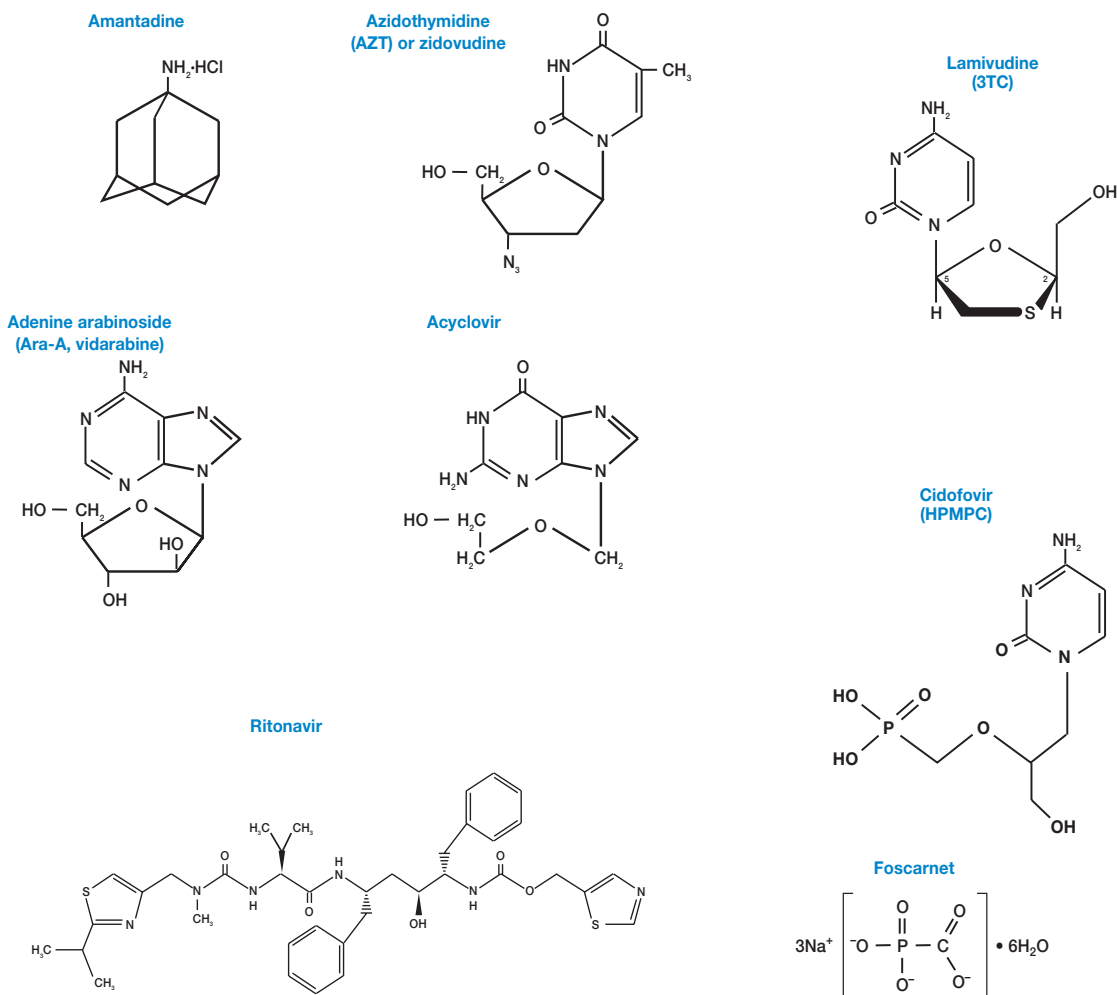
35.9 Antiviral Drugs

For many years the possibility of treating viral infections with drugs appeared remote because viruses enter host cells and make use of host cell enzymes and constituents to a large extent. A drug that would block virus reproduction also was thought to be toxic for the host. Inhibitors of virus-specific enzymes and life cycle processes have now been discovered, and several drugs are used therapeutically. Some important examples are shown in **figure 35.14**. [Animal virus life cycles \(pp. 399–409\)](#)

Most antiviral drugs disrupt either critical stages in the virus life cycle or the synthesis of virus-specific nucleic acids. **Aman-tadine** and rimantadine can be used to prevent influenza A infections. When given in time, it will reduce the incidence of influenza by 50 to 70% in an exposed population. Amantadine blocks the penetration and uncoating of influenza virus particles (*see section 18.2*). **Adenine arabinoside** or **vidarabine** disrupts the activity of DNA polymerase and several other enzymes involved in DNA and RNA synthesis and function. It is given intravenously or applied as an ointment to treat herpes infections. A

third drug, **acyclovir**, is also used in the treatment of herpes infections. Upon phosphorylation, acyclovir resembles deoxy-GTP and inhibits the virus DNA polymerase. Unfortunately acyclovir-resistant strains of herpes are already developing. Effective acyclovir derivatives and relatives are now available. Valacyclovir is an orally administered prodrug form of acyclovir. Ganciclovir, penciclovir, and its oral form famciclovir are effective in treatment of herpesviruses. Another kind of drug, foscarnet, inhibits the virus DNA polymerase in a different way. Foscarnet is an organic analogue of pyrophosphate (*figure 35.14*) that binds to the polymerase active site and blocks the cleavage of pyrophosphate from nucleoside triphosphate substrates. It is used in treating herpes and cytomegalovirus infections.

Several broad-spectrum anti-DNA virus drugs have been developed. A good example is the drug HPMPC or cidofovir (*figure 35.14*). It is effective against papovaviruses, adenoviruses, herpesviruses, iridoviruses, and poxviruses. The drug acts on the viral DNA polymerase as a competitive inhibitor and alternative substrate of dCTP. It has been used primarily against cytomegalovirus but also against herpes simplex and human papillomavirus infections.

**Figure 35.14** Representative Antiviral Drugs.

Research on anti-HIV drugs has been particularly active. Many of the first drugs to be developed were reverse transcriptase inhibitors such as **azidothymidine (AZT)** or **zidovudine**, lamivudine (3TC), didanosine (ddI), zalcitabine (ddC), and stavudine (d4T) (figure 35.14). These interfere with reverse transcriptase activity and therefore block HIV reproduction. More recently **HIV protease inhibitors** have been developed. Three of the most used are saquinvir, indinavir, and ritonavir (figure 35.14). These mimic the peptide bond that is normally attacked by the protease. The most successful treatment regimen involves a cocktail of agents given at high dosages to prevent the development of drug resistance. For example, the combination of AZT, 3TC, and ritonavir is very effective in reducing HIV plasma concentrations almost to zero. However, the treatment does not seem

able to eliminate latent proviral HIV DNA that still resides in memory T cells, and possibly elsewhere. [The biology and reproduction of HIV \(pp. 407, 878–84\)](#)

Probably the most publicized antiviral agents are **interferons**. These small proteins, produced by the host, inhibit virus replication and may be clinically useful in the treatment of influenza, hepatitis, herpes, and colds. The mechanism of interferon action is discussed in more detail in *section 31.9*.

1. Describe two different ways in which antiviral drugs disrupt virus reproduction and give an example of each.
2. Briefly summarize the way in which HIV is currently treated and how the two general classes of anti-HIV drugs act.

Summary

1. Chemotherapeutic agents are compounds that destroy pathogenic microorganisms or inhibit their growth and are used in the treatment of disease. Most are antibiotics: microbial products or their derivatives that can kill susceptible microorganisms or inhibit their growth.
2. The modern era of chemotherapy began with Paul Ehrlich's work on drugs against African sleeping sickness and syphilis. Other early pioneers were Gerhard Domagk, Alexander Fleming, Howard Florey, Ernst Chain, Norman Heatley, and Selman Waksman.
3. An effective chemotherapeutic agent must have selective toxicity. A drug with great selective toxicity has a high therapeutic index and usually disrupts a structure or process unique to the pathogen. It has fewer side effects.
4. Antibiotics can be classified in terms of the range of target microorganisms (narrow spectrum versus broad spectrum); their source (natural, semisynthetic, or synthetic); and their general effect (static versus cidal) (**table 35.1**).
5. Antibiotic effectiveness can be estimated through the determination of the minimal inhibitory concentration and the minimal lethal concentration with dilution susceptibility tests. Disk diffusion tests like the Kirby-Bauer test are often used to estimate a pathogen's susceptibility to drugs quickly (**figure 35.2**).
6. Chemotherapeutic agents can damage bacterial pathogens in several ways: inhibition of cell wall synthesis, protein synthesis, or nucleic acid synthesis; disruption of membrane structure; and inhibition of key enzymes (**table 35.4**).
7. A variety of factors can greatly influence the effectiveness of antimicrobial drugs during actual use.
8. Sulfonamides or sulfa drugs resemble *p*-aminobenzoic acid and competitively inhibit folic acid synthesis (**figure 35.3**).
9. Quinolones are a family of bactericidal synthetic drugs that inhibit DNA gyrase and thus inhibit such processes as DNA replication (**figure 35.6**).
10. Members of the penicillin family contain a β -lactam ring and disrupt bacterial cell wall synthesis, resulting in cell lysis (**figure 35.7**). Some, like penicillin G, are usually administered by injection and are most effective against gram-positive bacteria. Others can be given orally (penicillin V), are broad spectrum (ampicillin, carbenicillin), or are penicillinase resistant (methicillin).
11. Cephalosporins are similar to penicillins, and are given to patients with penicillin allergies (**figure 35.8**).
12. Tetracyclines are broad-spectrum antibiotics having a four-ring nucleus with attached groups (**figure 35.9**). They bind to the small ribosomal subunit and inhibit protein synthesis.
13. Aminoglycoside antibiotics like streptomycin and gentamicin bind to the small ribosomal subunit, inhibit protein synthesis, and are bactericidal (**figure 35.10**).
14. Erythromycin is a bacteriostatic macrolide antibiotic that binds to the large ribosomal subunit and inhibits protein synthesis (**figure 35.11**).
15. Vancomycin is a glycopeptide antibiotic that inhibits the transpeptidation reaction during peptidoglycan synthesis. It is used against drug-resistant staphylococci, enterococci, and clostridia.
16. Chloramphenicol is a broad-spectrum, bacteriostatic antibiotic that inhibits protein synthesis (**figure 35.12**). It is quite toxic and used only for very serious infections.
17. Bacteria can become resistant to a drug by excluding it from the cell, pumping the drug out of the cell, enzymatically altering it, modifying the target enzyme or organelle to make it less drug sensitive, and so forth. The genes for drug resistance may be found either on the bacterial chromosome or on a plasmid called an R plasmid.
18. Chemotherapeutic agent misuse fosters the increase and spread of drug resistance, and may lead to superinfections.
19. Because fungi are more similar to human cells than bacteria, antifungal drugs generally have a lower therapeutic index than antibacterial agents and produce more side effects.
20. Superficial mycoses can be treated with miconazole, ketoconazole, clotrimazole, tolnaftate, nystatin, and griseofulvin (**figure 35.13**). Amphotericin B, 5-flucytosine, and fluconazole are used for systemic mycoses.
21. Antiviral drugs interfere with critical stages in the virus life cycle (amantadine, rimantadine, and ritonavir) or inhibit the synthesis of virus-specific nucleic acids (zidovudine, adenine arabinoside, acyclovir) (**figure 35.14**). Interferon proteins inhibit virus replication and may be therapeutically useful in the future.

Key Terms

acyclovir 821
adenine arabinoside or vidarabine 821
amantadine 821
aminoglycoside antibiotic 816
amphotericin B 820
antibiotic 806
antimetabolites 812
azidothymidine (AZT) 822
broad-spectrum drugs 808
cephalosporin 814
chemotherapeutic agent 806
chloramphenicol 817
cidal 808
dilution susceptibility tests 809

erythromycin 817
griseofulvin 820
HIV protease inhibitors 822
interferon 822
Kirby-Bauer method 809
macrolide antibiotic 817
minimal inhibitory concentration (MIC) 809
minimal lethal concentration (MLC) 809
narrow-spectrum drugs 808
nystatin 820
parenteral route 812
penicillinase 814
penicillins 814
plasmids 819

quinolones 813
R plasmids 819
selective toxicity 807
side effects 808
static 808
streptomycin 816
structural analogues 812
sulfonamide 812
superinfection 819
tetracycline 815
therapeutic index 807
vancomycin 817
zidovudine 822

Questions for Thought and Review

1. Why do penicillins and cephalosporins have a much higher therapeutic index than most other antibiotics? What are antimetabolites?
2. Would there be any advantage to administering a bacteriostatic agent along with penicillin? Any disadvantage?
3. Why do antifungal drugs have a much lower therapeutic index than do most antibacterial drugs? Why does one often have to apply antifungal medications frequently?
4. What advantages would there be from administering two chemotherapeutic agents simultaneously?
5. Give several ways in which the development of antibiotic-resistant pathogens can be slowed or prevented.

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Critical Thinking Questions

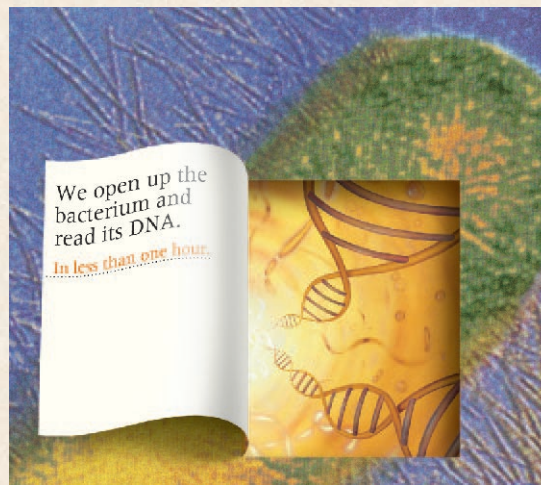
1. What advantage might soil bacteria and fungi gain from the synthesis of antibiotics?
2. Why might it be desirable to prepare a variety of semisynthetic antibiotics?
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CHAPTER 36

Clinical Microbiology



With microbial infections, time and accuracy in diagnosis are critically important. In clinical microbiology laboratories, new technologies (such as PCR and nucleic acid-based detection methods as shown in this illustration) are now being used with greater frequency to replace older methods of microbial diagnosis due to their increased accuracy and reduced time requirement.

Outline

- 36.1 Specimens 827
 - Collection 827
 - Handling 829
 - Transport 829
- 36.2 Identification of Microorganisms from Specimens 831
 - Microscopy 831
 - Growth and Biochemical Characteristics 831
 - Rapid Methods of Identification 840
 - Immunologic Techniques 842
 - Bacteriophage Typing 842
 - Molecular Methods and Analysis of Metabolic Products 842
- 36.3 Susceptibility Testing 844
- 36.4 Computers in Clinical Microbiology 844

Concepts

1. Clinical microbiologists and clinical microbiology laboratories perform many services, all related to the identification and control of microorganisms.
2. Success in clinical microbiology depends on (1) using the proper aseptic technique; (2) correctly obtaining the clinical specimen from the infected patient by swabs, needle aspiration, intubation, or catheters; (3) correctly handling the specimen; and (4) quickly transporting the specimen to the laboratory.
3. Once the clinical specimen reaches the laboratory, it is cultured and identified. Identification measures include microscopy; growth on enrichment, selective, differential, or characteristic media; specific biochemical tests; rapid test methods; immunologic techniques; bacteriophage typing; and molecular methods such as nucleic acid-based detection methods, gas-liquid chromatography, and plasmid fingerprinting.
4. After the microorganism has been isolated, cultured, and/or identified, samples are used in susceptibility tests to find which method of control will be most effective. The results are provided to the physician as quickly as possible.
5. Computer systems in clinical microbiology are designed to speed identification of the pathogen and communication of results back to the physician.

The specimen is the beginning. All diagnostic information from the laboratory depends upon the knowledge by which specimens are chosen and the care with which they are collected and transported.

—Cynthia A. Needham

Pathogens, particularly bacteria and yeasts, coexist with harmless microorganisms on or in the host. These pathogens must be properly identified as the actual cause of infectious diseases. This is the purpose of clinical microbiology. The clinical microbiologist identifies agents and organisms (hereafter referred to as microorganisms) based on morphological, biochemical, immunologic, and molecular procedures. Time is a significant factor in the identification process, especially in life-threatening situations. Computers and advances in technology for rapid identification, some commercially available, have greatly aided the clinical microbiologist. Molecular methods allow identification of microorganisms based on highly specific genomic and biochemical properties. Once isolated and identified, the microorganism can then be subjected to antimicrobial sensitivity tests. In the final analysis the patient's well-being and health can benefit significantly from information provided by the clinical microbiology laboratory—the subject of this chapter.

36.1 Specimens

The major focus of the **clinical microbiologist** is to isolate and identify microorganisms from clinical specimens rapidly. The purpose of the clinical microbiology laboratory is to provide the physician with information concerning the presence or absence of microorganisms that may be involved in the infectious disease process (**figure 36.1**). These individuals and facilities also determine the susceptibility of microorganisms to antimicrobial agents. Clinical microbiology makes use of information obtained from research on such diverse topics as microbial biochemistry and physiology, immunology, molecular biology, genomics, and the host-parasite relationships involved in the infectious disease process.

In clinical microbiology a clinical specimen (hereafter, specimen) represents a portion or quantity of human material that is tested, examined, or studied to determine the presence or absence of particular microorganisms. Safety for the patients, hospital, and laboratory staff is very important. The guidelines presented in **Box 36.1** (Universal Precautions for Health-Care Professionals) were established by the Centers for Disease Control and Prevention (CDC) to address areas of specimen handling. Other important concerns regarding specimens need emphasis:

1. The specimen selected should adequately represent the diseased area and also may include additional sites (e.g., liver and blood specimens) in order to isolate and identify potential agents of the particular disease process.
2. A quantity of specimen adequate in amount to allow a variety of diagnostic testing should be obtained.

3. Attention must be given to specimen collection in order to avoid contamination from the many varieties of microorganisms indigenous to the skin and mucous membranes (*see figure 31.2*).
4. The specimen should be forwarded promptly to the clinical laboratory.
5. If possible, the specimen should be obtained before antimicrobial agents have been administered to the patient.

Collection

Overall, the results obtained in the clinical laboratory are only as good as the quality of the specimen collected for analysis. Specimens may be collected by several methods using aseptic technique. Aseptic technique refers to specific procedures used to prevent unwanted microorganisms from contaminating the clinical specimen. Each method is designed to ensure that only the proper material will be sent to the clinical laboratory.

The most common method used to collect specimens from the anterior nares or throat is the sterile **swab**. A sterile swab is a rayon-, calcium alginate, or dacron-tipped polystyrene applicator. Manufacturers of swabs have their own unique container design and instructions for proper use. For example, many commercially manufactured swabs contain a transport medium designed to preserve a variety of microorganisms and to prevent multiplication of rapidly growing members of the population (**figure 36.2a**). However, with the exception of the nares or throat, the use of swabs for the collection of specimens is of little value and should be discouraged for two major reasons: swabs are associated with a greater risk of contamination with surface and subsurface microorganisms, and they have a limited volume capacity (<0.1 ml).

Needle aspiration is used to collect specimens aseptically (e.g., anaerobic bacteria) from cerebrospinal fluid, pus, and blood. For both samples stringent antiseptic techniques are used to avoid skin contamination. To prevent blood from clotting and entrapping microorganisms, various anticoagulants (e.g., heparin, sodium citrate) are included within the specimen bottle or tube (**figure 36.2b**).

Intubation [Latin *in*, into, and *tuba*, tube] is the inserting of a tube into a body canal or hollow organ. For example, intubation can be used to collect specimens from the stomach. In this procedure a long sterile tube is attached to a syringe, and the tube is either swallowed by the patient or passed through a nostril (**figure 36.2c**) into the patient's stomach. Specimens are then withdrawn periodically into the sterile syringe. The most common intubation tube is the Levin tube.

A **catheter** is a tubular instrument used for withdrawing or introducing fluids from or into a body cavity. For example, urine specimens may be collected with catheters to detect urinary tract infections caused by bacteria and from newborns and neonates who cannot give a voluntary urinary specimen. Three types are commonly used for urine. The hard catheter is used when the urethra is very narrow or has strictures. The French catheter is a soft tube used to obtain a single specimen sample. If multiple samples are required over a prolonged period, a Foley catheter is used (**figure 36.2d**).

The most common method used for the collection of urine is the clean-catch method. After the patient has cleansed the urethral



(a) The identification of the microorganism begins at the patient's bedside. The nurse is giving instructions to the patient on how to obtain a sputum specimen.



(b) The specimen is sent to the laboratory to be processed. Notice that the specimen and worksheet are in different Ziplock bags.



(c) Specimens such as sputum are plated on various types of media under a laminar airflow hood. This is to prevent specimen aerosols from coming in contact with the microbiologist.



(d) Sputum and other specimens are usually Gram stained to determine whether or not bacteria are present and to obtain preliminary results on the nature of any bacteria found.



(e) After incubation, the plates are examined for significant isolates. The Gram stain may be reexamined for correlation.



(f) Suspect colonies are picked for biochemical, immunologic, or molecular testing.



(g) Colonies are prepared for identification by rapid test systems.



(h) In a short period of time, sometimes 4 hours, computer-generated information is obtained that will consist of biochemical identification and antibiotic susceptibility results.



(i) All information about the specimen is now entered into a computer and the data are transmitted directly to the hospital ward.

Figure 36.1 Isolation and Identification of Microorganisms in a Clinical Laboratory.

Box 36.1

Universal Precautions for Health-Care Professionals

Since medical history and examination cannot reliably identify all patients infected with HIV or other blood-borne pathogens, blood and body-fluid precautions should be consistently used for all patients. Workers in microbiological research laboratories also are exposed to health hazards and need to employ universal precautions (*see Box 7.2, p. 145*).

1. All health-care workers should routinely use appropriate barrier precautions to prevent skin and mucous-membrane exposure when contact with blood or other body fluids of any patient is anticipated. Gloves should be worn for touching blood and body fluids, mucous membranes, or non-intact skin of all patients, for handling items or surfaces soiled with blood or body fluids, and for performing venipuncture and other vascular access procedures. Gloves should be changed after contact with each patient. Masks and protective eyewear or face shields should be worn during procedures that are likely to generate droplets of blood or other body fluids to prevent exposure of mucous membranes of the mouth, nose, and eyes. Gowns, aprons, or lab coats should be worn during procedures that are likely to generate splashes of blood or other body fluids.
2. Hands and other skin surfaces should be washed immediately and thoroughly if contaminated with blood or other body fluids. Hands should be washed immediately after gloves are removed.
3. All health-care workers should take precautions to prevent injuries caused by needles, scalpels, and other sharp instruments or devices during procedures; when cleaning used instruments; during disposal of used needles; and when handling sharp instruments after procedures. To prevent needlestick injuries, needles should not be recapped, purposely bent or broken by

hand, removed from disposable syringes, or otherwise manipulated by hand. After they are used, disposable syringes and needles, scalpel blades, and other sharp items should be placed in puncture-resistant containers for disposal.

4. Although saliva has not been implicated in HIV transmission, to minimize the need for emergency mouth-to-mouth resuscitation, mouthpieces, resuscitation bags, or other ventilation devices should be available for use in areas in which the need for resuscitation is predictable.
5. Health-care workers who have exudative lesions or weeping dermatitis should refrain from all direct patient care and from handling patient-care equipment.
6. The following procedure should be used to clean up spills of blood or blood-containing fluids. (1) Put on gloves and any other necessary barriers. (2) Wipe up excess material with disposable towels and place the towels in a container for sterilization. (3) Disinfect the area with either a commercial EPA-approved germicide or household bleach (sodium hypochlorite). The latter should be diluted from 1/100 (smooth surfaces) to 1/10 (porous or dirty surfaces); the dilution should be no more than 24 hours old. When dealing with large spills or those containing sharp objects such as broken glass, first cover the spill with disposable toweling. Then saturate the toweling with commercial germicide or a 1/10 bleach solution and allow it to stand for at least 10 minutes. Finally clean as described above.

Source: Adapted from Morbidity and Mortality Weekly Report, 36 (Suppl. 2S) 5S–10S. Centers for Disease Control and Prevention Guidelines, Atlanta, GA.

meatus (opening), a small container is used to collect the urine. The optimal time to use the clean-catch method is early morning because the urine contains more microorganisms as a result of being in the bladder overnight. In the clean-catch midstream method, the first urine voided is not collected because it will be contaminated with those transient microorganisms normally occurring in the lower portion of the urethra. Only the midstream portion is collected since it most likely will contain those microorganisms found in the urinary bladder. If warranted for some patients, needle aspirations also are done directly into the urinary bladder.

Sputum is the most common specimen collected in suspected cases of lower respiratory tract infections. Specifically, **sputum** is the mucous secretion expectorated from the lungs, bronchi, and trachea through the mouth, in contrast to saliva, which is the secretion of the salivary glands. Sputum is collected in specially designed sputum cups (figure 36.2e).

Handling

Immediately after collection the specimen must be properly labeled and handled. The person collecting the specimen is responsible for ensuring that the name, hospital, registration number, lo-

cation in the hospital, diagnosis, current antimicrobial therapy, name of attending physician, admission date, and type of specimen are correctly and legibly written or imprinted on the culture request form. This information must correspond to that written or imprinted on a label affixed to the specimen container. The type or source of the sample and the choice of tests to be performed also must be specified on the request form.

Transport

Speed in transporting the specimen to the clinical laboratory after it has been obtained from the patient is of prime importance. Some laboratories refuse to accept specimens if they have been in transit too long.

Microbiological specimens may be transported to the laboratory by various means (figure 36.1b). For example, certain specimens should be transported in a medium that preserves the microorganisms and helps maintain the ratio of one organism to another. This is especially important for specimens in which normal microorganisms may be mixed with microorganisms foreign to the body location.

Special treatment is required for specimens when the microorganism is thought to be anaerobic. The material is aspirated

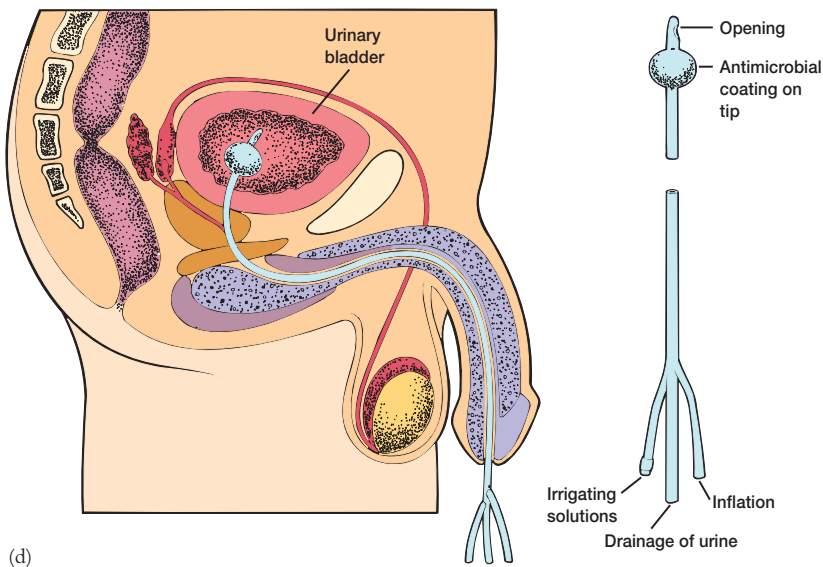
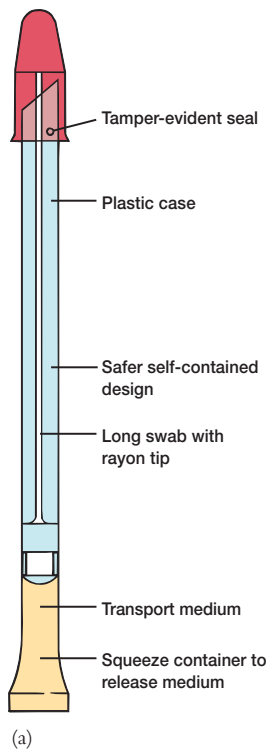


Figure 36.2 Collection of Clinical Specimens. (a) A drawing of a sterile swab with a specific transport medium. (b) Sterile Vacutainer tubes for the collection of blood. (c) Nasotracheal intubation. (d) A drawing of a Foley catheter. Notice that three separate lumens are incorporated within the round shaft of the catheter for drainage of urine, inflation, and introducing irrigating solutions into the urinary bladder. After the Foley catheter has been introduced into the urinary bladder, the tip is inflated to prevent it from being expelled. (e) This specially designed sputum cup allows the patient to expectorate a clinical specimen directly into the cup. In the laboratory, the cup can be opened from the bottom to reduce the chance of contamination from extraneous pathogens.



Figure 36.3 Some Anaerobic Transport Systems. A vial and syringe. These systems contain a nonnutritive transport medium that retards diffusion of oxygen after specimen addition and helps maintain microorganism viability up to 72 hours. A built-in color indicator system is clear and turns lavender in the presence of oxygen.

with a needle and syringe. Most of the time it is practical to remove the needle, cap the syringe with its original seal, and bring the specimen directly to the clinical laboratory. Transport of these specimens should take no more than 10 minutes; otherwise, the specimen must be injected immediately into an anaerobic transport vial (**figure 36.3**). Vials should contain a transport medium with an indicator, such as resazurin, to show that the interior of the vial is anaerobic at the time the specimen is introduced. Swabs for anaerobic culture usually are less satisfactory than aspirates or tissues, even if they are transported in an anaerobic vial.

Many clinical laboratories insist that stool specimens (the fecal discharge from the bowels) for culture be transported in various buffered preservatives. Preparation of these transport media is described in various manuals (see Additional Reading).

Transport of urine specimens to the clinical laboratory must be done as soon as possible. No more than 1 hour should elapse between the time the specimen is obtained and the time it is examined. If this time schedule cannot be followed, the urine sample must be refrigerated immediately.

Cerebrospinal fluid (CSF) from patients suspected of having meningitis should be examined immediately by skilled personnel in the clinical microbiology laboratory. CSF is obtained by lumbar puncture under conditions of strict asepsis, and the sample is transported to the laboratory within 15 minutes. Specimens for the isolation of viruses are iced before transport, and can be kept at 4°C for up to 72 hours; if the sample will be stored longer than 72 hours, it should be frozen at -72°C.

1. What is the function of the clinical microbiologist? The clinical microbiology laboratory?
2. What general guidelines should be followed in collecting and handling clinical specimens?
3. Define the following terms: specimen, swab, catheter, and sputum.
4. What are some transport problems associated with stool specimens? Anaerobic cultures? Urine specimens?

36.2 Identification of Microorganisms from Specimens

The clinical microbiology laboratory can provide preliminary or definitive identification of microorganisms based on (1) microscopic examination of specimens, (2) study of the growth and biochemical characteristics of isolated microorganisms (pure cultures), (3) immunologic tests that detect antibodies or microbial antigens, (4) bacteriophage typing (restricted to research settings and the CDC), and (5) molecular methods.

Microscopy

Wet-mount, heat-fixed, or chemically fixed specimens can be examined with an ordinary bright-field microscope. These preparations can be enhanced with either phase-contrast or dark-field microscopy. The latter is the procedure of choice for the detection of spirochetes in skin lesions associated with early syphilis or in blood specimens of people with early leptospirosis. The fluorescence microscope can be used to identify certain acid-fast microorganisms (*Mycobacterium tuberculosis*) after they are stained with fluorochromes such as auramine-rhodamine (see section 2.2). (Some morphological features used in classification and identification of microorganisms are presented in section 19.5 and in table 19.3.) [The light microscope \(pp. 19–27\).](#)

Many stains that can be used to examine specimens for specific microorganisms have been described. Two of the more widely used are the Gram stain and acid-fast stain. Because these stains are based on the chemical composition of cell walls, they are not useful in identifying bacteria without walls. Refer to standard references, such as the *Manual of Clinical Microbiology* published by the American Society for Microbiology, for details about other reagents and staining procedures.

Growth and Biochemical Characteristics

Typically microorganisms have been identified by their particular growth patterns and biochemical characteristics. These characteristics vary depending on whether the clinical microbiologist is dealing with viruses, rickettsias, chlamydiae, mycoplasmas, gram-positive or gram-negative bacteria, fungi (yeasts, molds), or parasites (protozoa, helminths).

Viruses

Viruses are identified by isolation in conventional cell (tissue) culture, by immunodiagnosis (fluorescent antibody, enzyme immunoassay,

radioimmunoassay, latex agglutination, and immunoperoxidase) tests, and by molecular detection methods such as nucleic acid probes and amplification assays. Several types of systems are available for virus cultivation: cell cultures, embryonated hen's eggs, and experimental animals. [Immunological tests \(section 33.3\)](#)

Cell cultures are divided into three general classes:

1. Primary cultures consist of cells derived directly from tissues such as monkey kidney and mink lung cells that have undergone one or two passages since harvesting.
2. Semicontinuous cell cultures or low-passage cell lines are obtained from subcultures of a primary culture and usually consist of diploid fibroblasts that undergo a finite number of divisions.
3. Continuous cell cultures, such as HEP-2 cells, are derived from transformed cells that are generally epithelial in origin. These cultures grow rapidly, are heteroploid (having a chromosome number that is not a simple multiple of the haploid number), and can be subcultured indefinitely.

Each type of cell culture favors the growth of a different array of viruses, just as bacterial culture media have differing selective and restrictive properties for growth of bacteria.

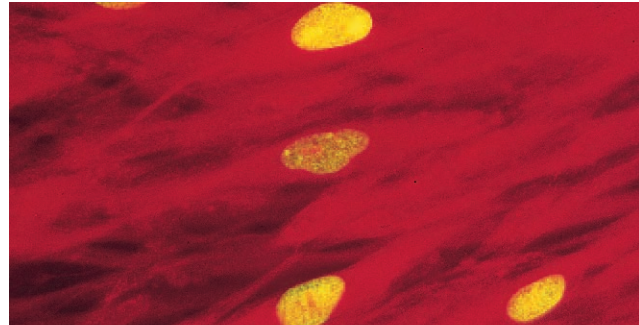
Viral replication in cell cultures is detected in two ways: (1) by observing the presence or absence of cytopathic effects (CPEs), and (2) by hemadsorption.

A **cytopathic effect** is an observable morphological change that occurs in cells because of viral replication (*see section 16.3*). Examples include ballooning, binding together, clustering, or even death of the culture cells (*see figure 16.3*). During the incubation period of a cell culture, red blood cells can be added. Several viruses alter the plasma membrane of infected culture cells so that red blood cells adhere firmly to them. This phenomenon is called **hemadsorption** (*see figure 33.10*).

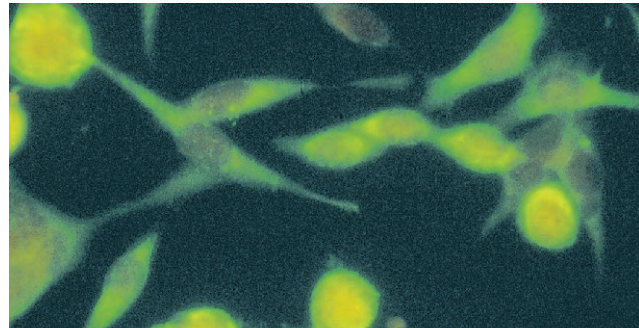
Embryonated hen's eggs can be used for virus isolation. There are three main routes of egg inoculation for virus isolation: (1) the allantoic cavity, (2) the amniotic cavity, and (3) the chorioallantoic membrane (*see figure 16.1*). Virus replication is recognized by the development of pocks on the chorioallantoic membrane, by the development of hemagglutinins (*see figure 33.10*) in the allantoic and amniotic fluid, and by death of the embryo.

Laboratory animals, especially suckling mice, are used for virus isolation. Inoculated animals are observed for specific signs of disease or death.

Several new serological tests for viral identification make use of monoclonal antibody-based immunofluorescence. These tests (**figure 36.4**) detect viruses such as the cytomegalovirus and herpes simplex virus in tissue-vial cultures.



(a)



(b)

Figure 36.4 Viral Identification Test Using Immunofluorescence in Tissue Culture. (a) Two infected nuclei in a cytomegalovirus (CMV) positive tissue culture. (b) Several infected cells in a herpes simplex virus positive tissue culture.

Fungi

Fungal infections (i.e., mold and yeast infections) often are diagnosed by direct microscopic (fluorescence) examination of specimens. For example, the identification of molds often can be made if a portion of the specimen is mixed with a drop of 10% Calcofluor White stain on a glass slide. Fungal cultures remain as the standard for the recovery of fungi from patient specimens; however, the time needed to culture fungi varies anywhere from a few days to several weeks depending on the organism. Fungal serology (e.g., complement fixation and immunodiffusion) is designed to detect serum antibody but is limited to a few fungi (*Blastomyces dermatitidis*, *Coccidioides immitis*, *Histoplasma capsulatum*). The cryptococcal latex antigen test is routinely used for the direct detection of *Cryptococcus neoformans* in serum and cerebrospinal fluid. In the clinical laboratory, nonautomated (conventional kits) and automated methods for rapid identification (4 to 24 hours) are used to detect most yeasts. Any biochemical methods used to detect fungi should always be accompanied by morphological studies examining for pseudohyphae, yeast cell structure, chlamydospores, and so on.

1. Name two specimens for which microscopy would be used in the initial diagnosis of an infectious disease.
2. Name three general classes of cell cultures.
3. Give two ways by which the presence of viral replication is detected in cell culture.
4. What are the three main routes of egg inoculation for virus isolation?

Table 36.1 Isolation of Pure Bacterial Cultures from Specimens**Selective Media**

A selective medium is prepared by the addition of specific substances to a culture medium that will permit growth of one group of bacteria while inhibiting growth of some other groups. The following are examples:

Salmonella-Shigella agar (SS) is used to isolate *Salmonella* and *Shigella* species. Its bile salt mixture inhibits many groups of coliforms. Both *Salmonella* and *Shigella* species produce colorless colonies because they are unable to ferment lactose. Lactose-fermenting bacteria will produce pink colonies.

Mannitol salt agar (MS) is used for the isolation of staphylococci. The selectivity is obtained by the high (7.5%) salt concentration that inhibits growth of many groups of bacteria. The mannitol in this medium helps in differentiating the pathogenic from the nonpathogenic staphylococci, as the former ferment mannitol to form acid while the latter do not.

Bismuth sulfite agar (BS) is used for the isolation of *Salmonella typhi*, especially from stool and food specimens. *S. typhi* reduces the sulfite to sulfide, resulting in black colonies with a metallic sheen.

Differential Media

The incorporation of certain chemicals into a medium may result in diagnostically useful growth or visible change in the medium after incubation. The following are examples:

Eosin methylene blue agar (EMB) differentiates between lactose fermenters and nonlactose fermenters. EMB contains lactose, salts, and two dyes—eosin and methylene blue. *E. coli*, which is a lactose fermenter, will produce a dark colony or one that has a metallic sheen. *S. typhi*, a nonlactose fermenter, will appear colorless.

MacConkey agar is used for the selection and recovery of *Enterobacteriaceae* and related gram-negative rods. The bile salts and crystal violet in this medium inhibit the growth of gram-positive bacteria and some fastidious gram-negative bacteria. Because lactose is the sole carbohydrate, lactose-fermenting bacteria produce colonies that are various shades of red, whereas nonlactose fermenters produce colorless colonies.

Hektoen enteric agar is used to increase the yield of *Salmonella* and *Shigella* species relative to other microbiota. The high bile salt concentration inhibits the growth of gram-positive bacteria and retards the growth of many coliform strains.

Enrichment Media

The addition of blood, serum, or extracts to tryptic soy agar or broth will support the growth of many fastidious bacteria. These media are used primarily to isolate bacteria from cerebrospinal fluid, pleural fluid, sputum, and wound abscesses. The following are examples:

Blood agar (can also be a differential medium): addition of citrated blood to tryptic soy agar makes possible variable hemolysis, which permits differentiation of some species of bacteria. Three hemolytic patterns can be observed on blood agar.

1. α -hemolysis—greenish to brownish halo around the colony (e.g., *Streptococcus gordonii*, *Streptococcus pneumoniae*).
2. β -hemolysis—complete lysis of blood cells resulting in a clearing effect around growth of the colony (e.g., *Staphylococcus aureus* and *Streptococcus pyogenes*).
3. Nonhemolytic—no change in medium (e.g., *Staphylococcus epidermidis* and *Staphylococcus saprophyticus*).

Chocolate agar is made from heated blood, which provides necessary growth factors to support bacteria such as *Haemophilus influenzae* and *Neisseria gonorrhoeae*.

Characteristic Media

Characteristic media are used to test bacteria for particular metabolic activities, products, or requirements. The following are examples:

Urea broth is used to detect the enzyme urease. Some enteric bacteria are able to break down urea, using urease, into ammonia and CO₂.

Triple sugar iron (TSI) agar contains lactose, sucrose, and glucose plus ferrous ammonium sulfate and sodium thiosulfate. TSI is used for the identification of enteric organisms based on their ability to attack glucose, lactose, or sucrose and to liberate sulfides from ammonium sulfate or sodium thiosulfate.

Citrate agar contains sodium citrate, which serves as the sole source of carbon, and ammonium phosphate, the sole source of nitrogen. Citrate agar is used to differentiate enteric bacteria on the basis of citrate utilization.

Lysine iron agar (LIA) is used to differentiate bacteria that can either deaminate or decarboxylate the amino acid lysine. LIA contains lysine, which permits enzyme detection, and ferric ammonium citrate for the detection of H₂S production.

Sulfide, indole, motility (SIM) medium is used for three different tests. One can observe the production of sulfides, formation of indole (a metabolic product from tryptophan utilization), and motility. This medium is generally used for the differentiation of enteric organisms.

Parasites

Concentrated wet mounts of blood, stool, or urine specimens can be examined microscopically for the presence of eggs, cysts, larvae, or vegetative cells of parasites. Blood smears for sporozoan (malaria) and flagellate (trypanosome) parasites are stained with Giemsa. Some serological tests also are available.

Bacteria

Isolation and growth of bacteria are required before many diagnostic tests can be used to confirm the identification of the pathogen. The presence of bacterial growth usually can be recog-

nized by the development of colonies on solid media or turbidity in liquid media. The time for visible growth to occur is an important variable in the clinical laboratory. For example, most pathogenic bacteria require only a few hours to produce visible growth, whereas it may take weeks for colonies of mycobacteria or mycoplasmas to become evident. The clinical microbiologist as well as the clinician should be aware of reasonable reporting times for various cultures.

The initial identity of a bacterial organism may be suggested by (1) the source of the culture specimen; (2) its microscopic appearance and Gram stain; (3) its pattern of growth on selective, differential, enrichment, or characteristic media (**table 36.1**; *see also figure 5.11*); and (4) its hemolytic, metabolic, and fermentative

Table 36.2 Some Biochemical Tests Used by Clinical Microbiologists in the Diagnosis of Bacteria from the Patient's Specimen

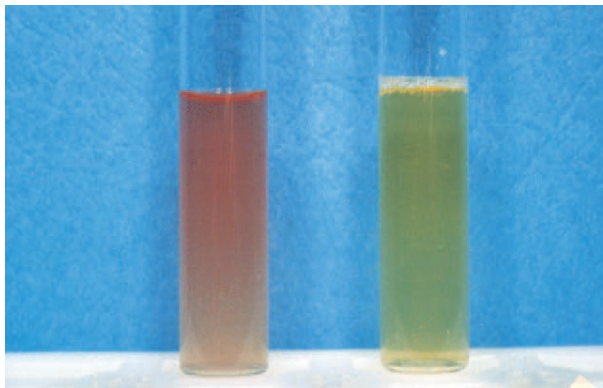
Biochemical Test	Description	Laboratory Application
Carbohydrate fermentation (figure 36.5m)	Acid and/or gas are produced during fermentative growth with sugars or sugar alcohols.	Fermentation of specific sugars used to differentiate enteric bacteria as well as other genera or species.
Casein hydrolysis	Detects the presence of caseinase, an enzyme able to hydrolyze milk protein casein. Bacteria that use casein appear as colonies surrounded by a clear zone.	Used to cultivate and differentiate aerobic actinomycetes based on casein utilization. For example, <i>Streptomyces</i> uses casein and <i>Nocardia</i> does not.
Catalase (figure 36.5d,e)	Detects the presence of catalase, which converts hydrogen peroxide to water and O ₂ .	Used to differentiate <i>Streptococcus</i> (–) from <i>Staphylococcus</i> (+) and <i>Bacillus</i> (+) from <i>Clostridium</i> (–).
Citrate utilization	When citrate is used as the sole carbon source, this results in alkalization of the medium	Used in the classification of enteric bacteria. <i>Klebsiella</i> (+), <i>Enterobacter</i> (+), <i>Salmonella</i> (often +); <i>Escherichia</i> (–), <i>Edwardsiella</i> (–).
Coagulase	Detects the presence of coagulase. Coagulase causes plasma to clot.	This is an important test to differentiate <i>Staphylococcus aureus</i> (+) from <i>S. epidermidis</i> (–).
Decarboxylases (arginine, lysine, ornithine) (figure 36.5g)	The decarboxylation of amino acids releases CO ₂ and amine.	Used in the classification of enteric bacteria.
Esculin hydrolysis	Tests for the cleavage of a glycoside.	Used in the differentiation of <i>Staphylococcus aureus</i> , <i>Streptococcus mitis</i> , and others (–) from <i>S. bovis</i> , <i>S. mutans</i> , and enterococci (+).
β-galactosidase (ONPG) test	Demonstrates the presence of an enzyme that cleaves lactose to glucose + galactose.	Used to separate enterics (<i>Citrobacter</i> +, <i>Salmonella</i> –) and to identify pseudomonads.
Gelatin liquefaction (figure 36.5i)	Detects whether or not a bacterium can produce proteases that hydrolyze gelatin and liquify solid gelatin medium.	Used in the identification of <i>Clostridium</i> , <i>Serratia</i> , <i>Pseudomonas</i> , and <i>Flavobacterium</i> .
Hydrogen sulfide (H ₂ S)	Detects the formation of hydrogen sulfide from the amino acid cysteine due to cysteine desulfurase.	Important in the identification of <i>Edwardsiella</i> , <i>Proteus</i> , and <i>Salmonella</i> .
IMViC (indole; methyl red; Voges-Proskauer; citrate) (figure 36.5a,b,h)	The indole test detects the production of indole from the amino acid tryptophan. Methyl red is a pH indicator to determine whether the bacterium has produced acid. VP (Voges-Proskauer) detects the production of acetoin. The citrate test determines whether or not the bacterium can use sodium citrate as a sole source of carbon.	Used to separate <i>Escherichia</i> (MR+, VP–, indole+) from <i>Enterobacter</i> (MR–, VP+, indole–) and <i>Klebsiella pneumoniae</i> (MR–, VP+, indole–); also used to characterize members of the genus <i>Bacillus</i> .
Lipid hydrolysis	Detects the presence of lipase, which breaks down lipids into simple fatty acids and glycerol.	Used in the separation of clostridia.
Nitrate reduction (figure 36.5f)	Detects whether a bacterium can use nitrate as an electron acceptor.	Used in the identification of enteric bacteria which are usually +.
Oxidase (figure 36.5n,p)	Detects the presence of cytochrome <i>c</i> oxidase that is able to reduce O ₂ and artificial electron acceptors.	Important in distinguishing <i>Neisseria</i> and <i>Moraxella</i> spp. (+) from <i>Acinetobacter</i> (–), and enterics (all –) from pseudomonads (+).
Phenylalanine deaminase (figure 36.5j)	Deamination of phenylalanine produces phenylpyruvic acid, which can be detected colorimetrically.	Used in the characterization of the genera <i>Proteus</i> and <i>Providencia</i> .
Starch hydrolysis (figure 36.5c)	Detects the presence of the enzyme amylase, which hydrolyzes starch.	Used to identify typical starch hydrolyzers such as <i>Bacillus</i> spp.
Urease (figure 36.5r)	Detects the enzyme that splits urea to NH ₃ and CO ₂	Used to distinguish <i>Proteus</i> , <i>Providencia rettgeri</i> , and <i>Klebsiella pneumoniae</i> (+) from <i>Salmonella</i> , <i>Shigella</i> and <i>Escherichia</i> (–).

properties on the various media (table 36.1; *see also table 19.4*). After the microscopic and growth characteristics of a pure culture of bacteria are examined, specific biochemical tests can be performed. Some of the most common biochemical tests used to identify bacterial isolates are given in **table 36.2** and **figure 36.5**. **Microbial nutrition and types of media** (chapter 5)

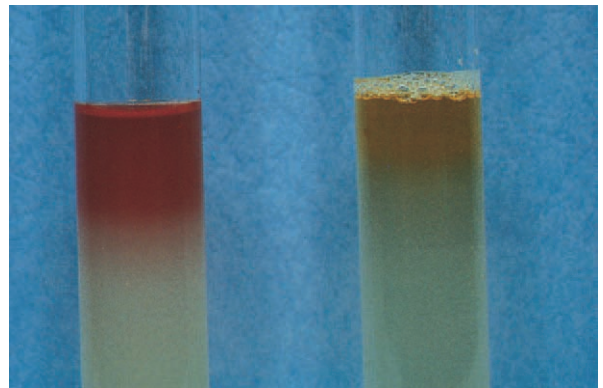
Classic dichotomous keys are coupled with the biochemical tests for the identification of bacteria from specimens. Generally, fewer than 20 tests are required to identify clinical bacterial isolates to the species level (**figure 36.6**).

Rickettsias

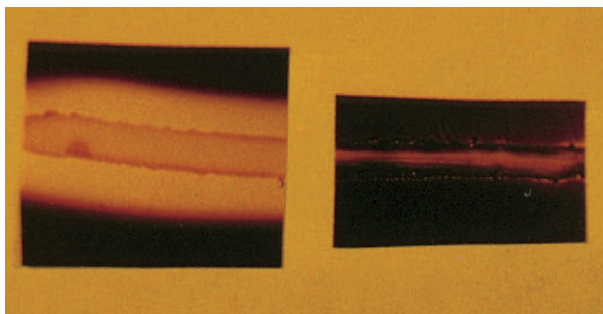
Although rickettsias, chlamydiae, and mycoplasmas are bacteria, they differ from other bacterial pathogens in a variety of ways. Therefore the identification of these three groups is discussed separately. Rickettsias can be diagnosed by immunoassays or by isolation of the microorganism. Because isolation is both hazardous to the clinical microbiologist and expensive, immunological methods are preferred. Isolation of rickettsias and diagnosis of rickettsial diseases is generally confined to reference and specialized research laboratories.



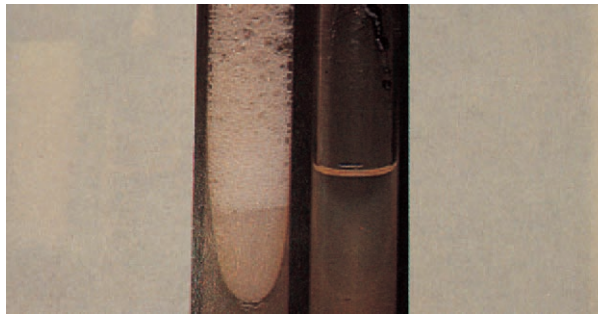
(a) Methyl Red Test. This is a qualitative test of the acidity produced by bacteria grown in MR-VP broth. After the addition of several drops of methyl red solution, a bright red color (left tube) is a positive test; a yellow or orange color (right tube) is a negative test. Used to separate enterics, for example *E. coli* is MR positive and *Enterobacter aerogenes* is MR negative.



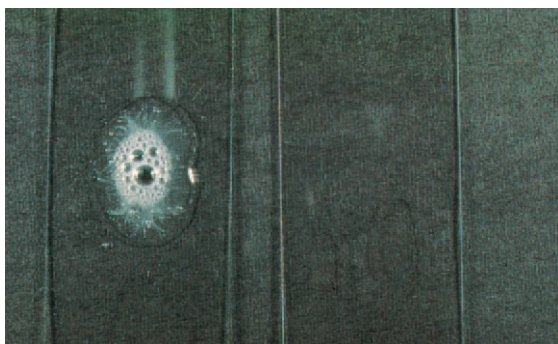
(b) Voges-Proskauer Reaction. VP-positive bacteria produce acetylmethylcarbinol or acetoin, which reacts with the reagents to produce a red color (left tube); a VP-negative tube is shown on the right. Used to differentiate *Bacillus* species and enterics. For example, *E. coli* is VP negative and *Enterobacter aerogenes* is VP positive.



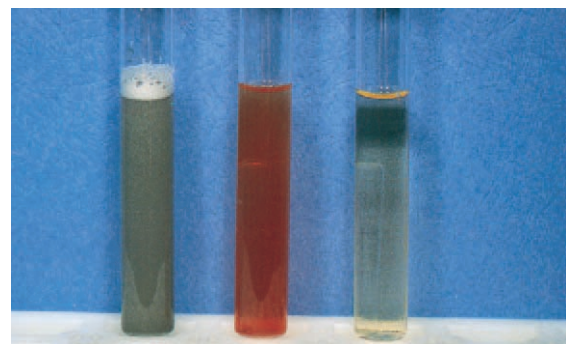
(c) Starch Hydrolysis. After incubation on starch agar, plates are flooded with iodine solution. A positive test is indicated by the colorless area around the growth (left); a negative test is shown on the right. Used to detect the production of α -amylase by certain bacteria.



(d) Tube Catalase Test. After incubation of slant cultures, 1 ml of 3% hydrogen peroxide is trickled down the slants. Catalase converts hydrogen peroxide to water and oxygen bubbles (left tube); a negative catalase test is shown in the right tube. Used to differentiate *Streptococcus* (catalase negative) from *Staphylococcus* (catalase positive).

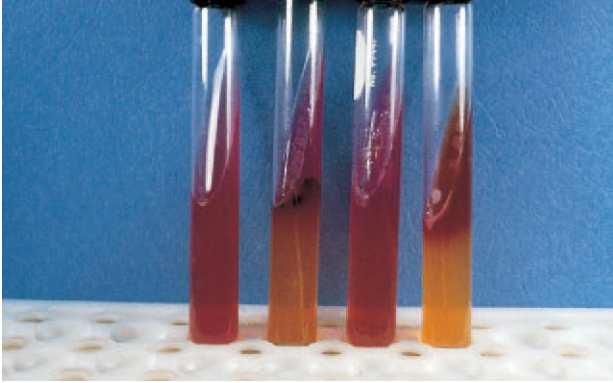


(e) Slide Catalase Test. A wooden applicator stick (or nichrome wire loop) is used to pick up a colony from a culture plate and place it in a drop of hydrogen peroxide on a glass slide. A positive catalase reaction (left slide) shows gas bubbles; a negative catalase reaction reveals an absence of gas bubbles (right slide). Used to differentiate *Streptococcus* (catalase negative) from *Staphylococcus* (catalase positive).

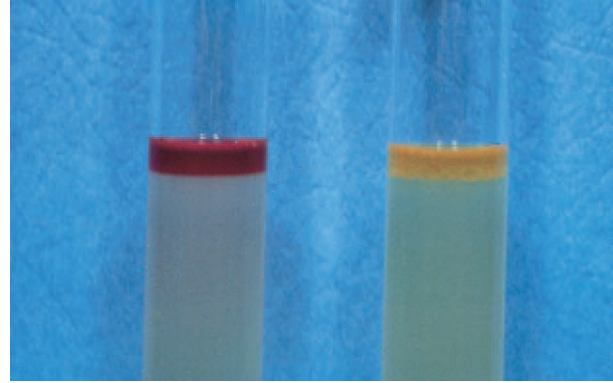


(f) Nitrate Reduction. After 24–48 hours of incubation, nitrate reagents are added to culture tubes. The tube on the left illustrates gas formation (a positive reaction for nitrate reduction); the tube in the middle is a positive reaction for nitrate reduction to nitrite as indicated by the red color; the tube on the right is a negative broth control. Used to test for miscellaneous gram-negative bacteria.

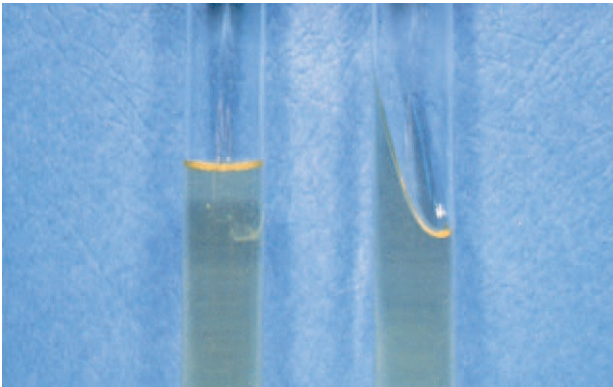
Figure 36.5 Some Common Diagnostic Tests Used in Microbiology.



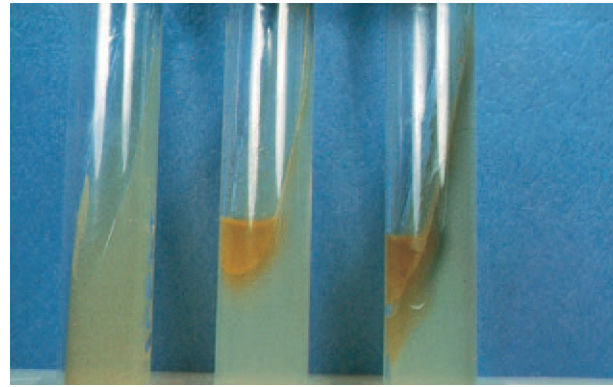
(g) **Test for Amino Acid Decarboxylase.** The tube on the left is an uninoculated control; the second tube from the left is lysine decarboxylase negative; the third tube is lysine decarboxylase positive; and the tube on the right is lysine deaminase positive. Used to classify enteric bacteria. Recommended for detection of arginine dihydrolase and lysine and ornithine decarboxylase activities.



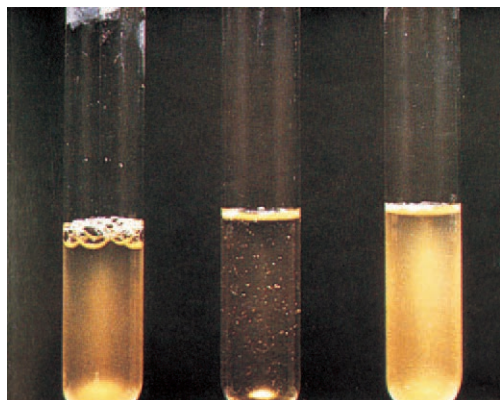
(h) **Test for Indole.** Tryptophan can be broken down to indole by some bacteria. The presence of indole is detected by adding Kovacs' reagent. A red color on the surface is a positive test for indole (left tube) and an orange-yellow color is a negative test for indole (right tube). Used to separate enterics such as *E. coli* (indole positive) and *Enterobacter* (indole negative).



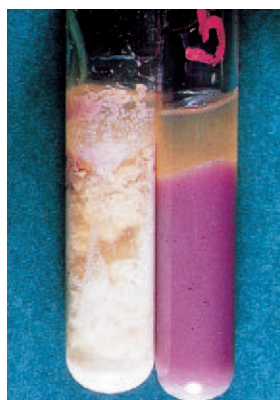
(i) **Gelatin Hydrolysis or Liquefaction.** If gelatin is hydrolyzed by the enzyme gelatinase, it does not gel when cooled but remains a liquid. Thus it flows when the culture is tilted backward (right tube). A negative control is on the left. Note that the solid gelatin does not flow when the tube is tilted. Used to differentiate a variety of heterotrophic bacteria, such as clostridia.



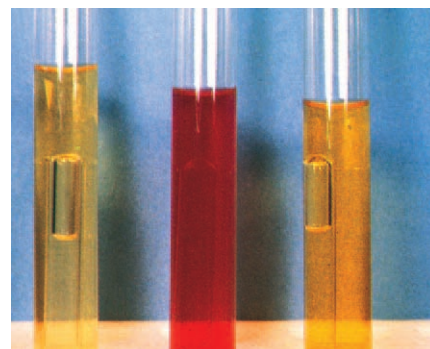
(j) **Phenylalanine Deamination Test.** When 10% ferric chloride is added to a phenylalanine deaminase agar slant culture, a dark green color (tube on the right) is a positive test for the enzyme. The tube on the left is an uninoculated control, and the tube in the middle is a negative test. Used to differentiate members of the *Proteus* group (*Proteus vulgaris*, positive, from *E. coli*, negative).



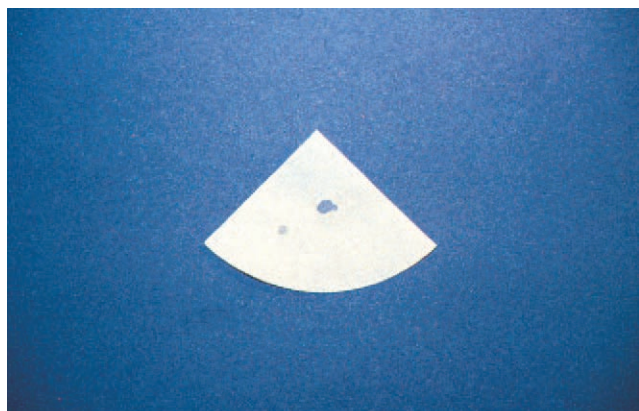
(k) Bile Solubility for Pneumococcus. The tube on the left and the one in the middle contain cultures of pneumococcus (*Streptococcus pneumoniae*). The tube on the right contains α -streptococcus. One-tenth ml of deoxycholate was added to the middle and right tubes, and 0.1 ml of distilled water was added to the left tube. The pneumococci in the center are bile soluble, as indicated by the clear suspension. The bacteria in the right tube are not bile soluble, as indicated by the turbidity.



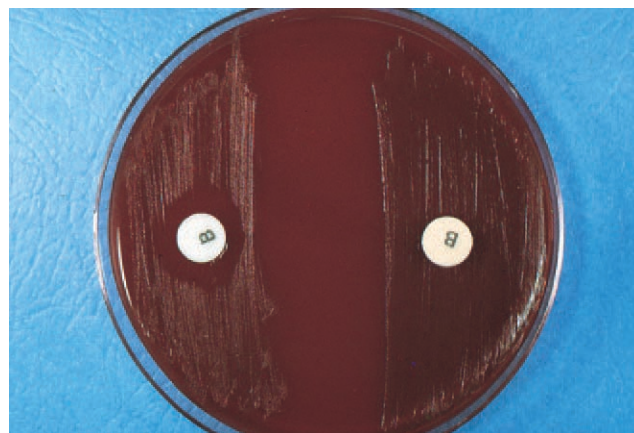
(l) Stormy Fermentation of Litmus Milk. The tube on the left shows fermentation; the tube on the right is negative for stormy fermentation. Used for the identification of *Clostridium* species.



(m) Fermentation Reactions. The tube on the left shows acid (yellow color) and gas in the Durham tube. The center tube shows no carbohydrate utilization to produce acid or gas. The tube on the right has less acid formation than that on the left. Used to separate enterics.

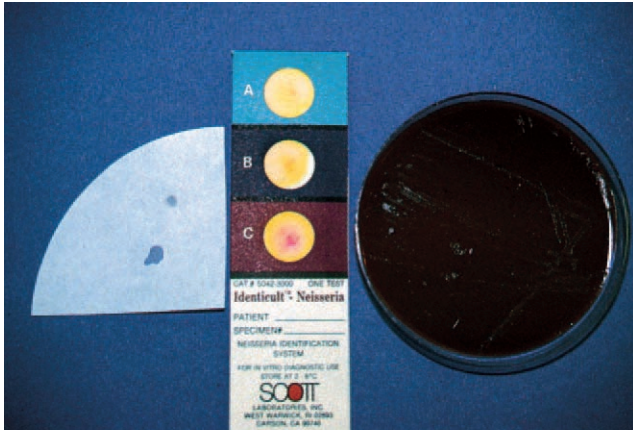


(n) Oxidase Test. Filter paper is moistened with a few drops of 1% tetramethyl-*p*-phenylenediamine dihydrochloride. With a wooden applicator, growth from an agar medium is smeared on the paper. A positive test is the development of a purple color within 10 seconds. Used to separate enterics from pseudomonads. For example, *Pseudomonas aeruginosa* is positive and *E. coli* is negative.

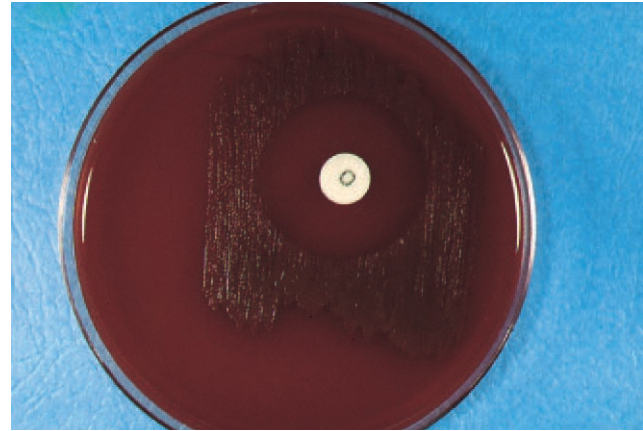


(o) Bacitracin Sensitivity Test. The bacteria on the left are presumptively identified as group A streptococci because of inhibition by the antibiotic bacitracin. The bacteria on the right (*Enterococcus faecalis*) are bacitracin resistant.

Figure 36.5 continued



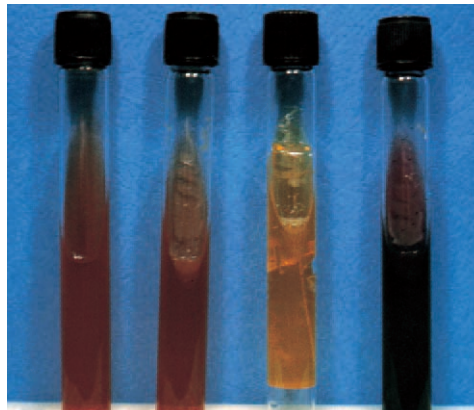
(p) **Identification of *Neisseria gonorrhoeae*.** Oxidase-positive test on filter paper is shown on the left. Identicult reaction for *N. gonorrhoeae* (center). The plate on the right shows characteristic growth on modified Thayer-Martin medium.



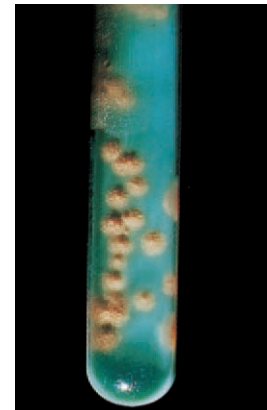
(q) **Optochin Sensitivity.** An optochin disk on blood agar specifically inhibits pneumococci. The optochin test for *Streptococcus pneumoniae* is based on the zone of inhibition.



(r) **Urease Production.** Christensen urea agar slants. Urease-positive bacteria hydrolyze urea to ammonia, which turns the phenol red indicator red-violet. From left to right: uninoculated control, delayed positive (>24 hrs), rapidly positive (<4 hrs), negative reaction. Used to separate enterics.



(s) **Triple Sugar Iron Agar Reactions.** Left to right: left tube is an uninoculated control, second tube is K/K (nonfermenter), third tube is A/A with gas indicating lactose or sucrose fermentation, and the right tube is K/A plus H₂S production. A stands for acid production, and K indicates that the medium becomes alkaline. Used for the differentiation of members of the *Enterobacteriaceae* based on the fermentation of lactose, sucrose, glucose, and the production of H₂S.



(t) **Lowenstein-Jensen Medium.** Growth of *Mycobacterium tuberculosis* showing nodular and nonpigmented growth.

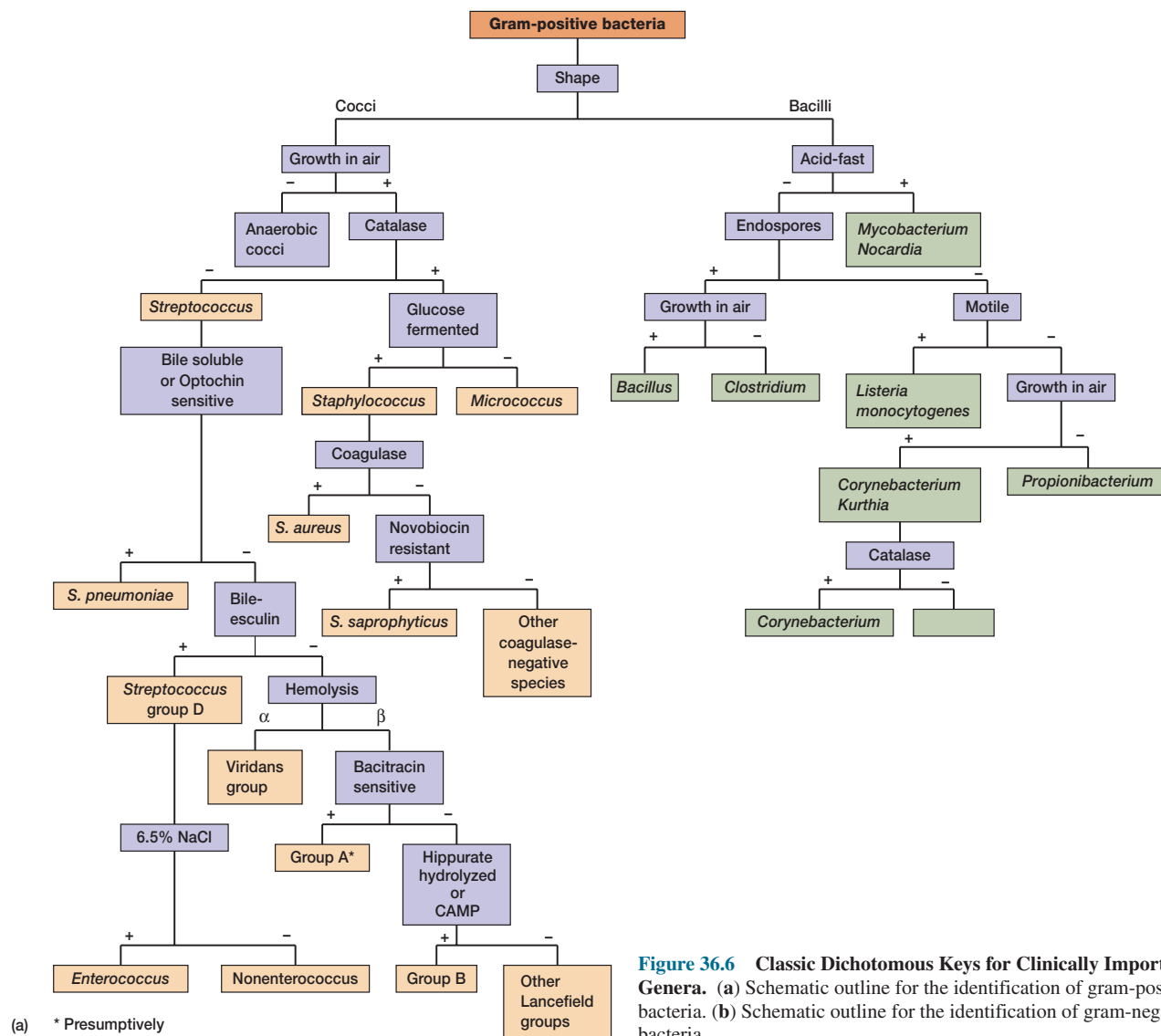


Figure 36.6 Classic Dichotomous Keys for Clinically Important Genera. (a) Schematic outline for the identification of gram-positive bacteria. (b) Schematic outline for the identification of gram-negative bacteria.

Chlamydiae

Chlamydiae can be demonstrated in tissues and cell scrapings with Giemsa staining, which detects the characteristic intracellular inclusion bodies (see figure 21.13). Immunofluorescent staining of tissues and cells with monoclonal antibody reagents is a more sensitive and specific means of diagnosis (Box 36.2; see also figure 32.20). The most sensitive methods for demonstrating chlamydiae in clinical specimens are DNA probes (see p. 322) and PCR methods (see section 14.3).

Mycoplasmas

The most routinely used techniques for identification of the mycoplasmas are immunological (hemagglutinin) or complement-fixing antigen-antibody reactions using the patient's sera. These

microorganisms are slow growing; therefore positive results from isolation procedures are rarely available before 30 days—a long delay with an approach that offers little advantage over standard techniques. Recently DNA probes have been applied to the detection of *Mycoplasma pneumoniae* in clinical specimens.

1. How can fungi and parasites be detected in a clinical specimen? Rickettsias? Chlamydiae? Mycoplasmas?
2. Why must the clinical microbiologist know what are reasonable reporting times for various microbial specimens?
3. How can a clinical microbiologist determine the initial identity of a bacterium?
4. Describe a dichotomous key that could be used to identify a bacterium.

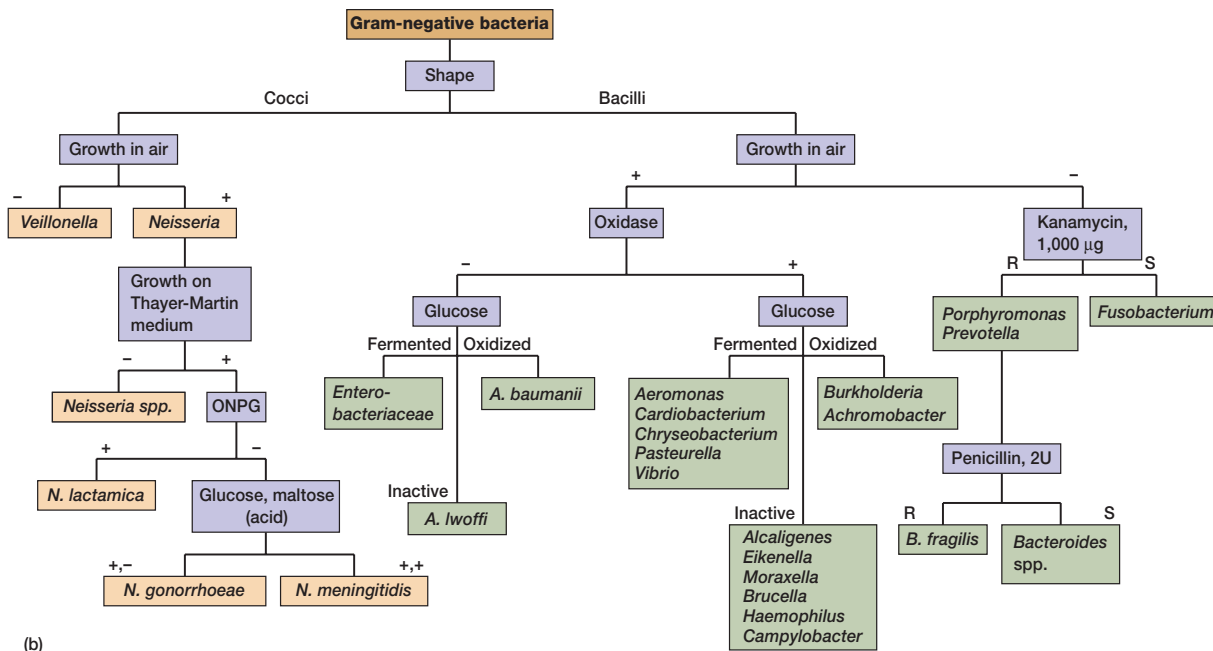


Figure 36.6 continued

Box 36.2

Monoclonal Antibodies in Clinical Microbiology

An important application of monoclonal antibody technology (see *hybridomas*, section 32.3) is the identification of microorganisms. Monoclonal antibodies have been prepared for a wide range of viruses, bacteria, fungi, and parasites; however, many remain as research tools and are not commercially available. If specific monoclonal antibodies are selected, immunologic assays can be created for different types of analyses. For example, monoclonal antibodies of cross-species or cross-genus reactivity have applications in the taxonomy of microorganisms. Those monoclonal antibodies that define species-specific antigens are extremely valuable in diagnostic reagents. Monoclonal antibodies that exhibit more restrictive specificity can be used to identify strains or biotypes within a species, to aid in studies of antigenic drift, and in epidemiological studies involving the matching of microbial strains. In addition, individual antigenic determinants on protein molecules can be mapped.

In the clinical microbiology laboratory, monoclonal antibodies to viral or bacterial antigens are replacing polyclonal antibodies for use in culture confirmation when accurate, rapid identification is required. With the use of sensitive techniques such as fluorescent antibody assays it is possible to perform culture identifications with improved accuracy, speed, and fewer microorganisms. The formulation of direct assays with monoclonal antibody reagents, which contain no contaminating antibodies and produce a minimum of artifacts, is now reality. The highly defined and reproducible properties of monoclonal antibodies invite their incorporation into immunoassays being developed for the next generation of instruments that will detect microbial antigens and serum antibodies for the clinical microbiologist.

Rapid Methods of Identification

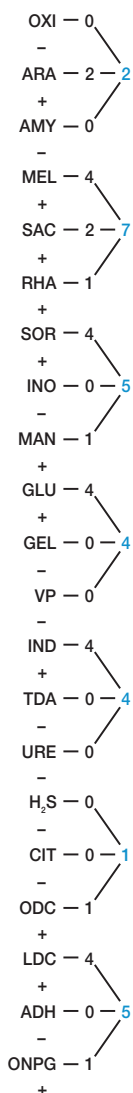
Clinical microbiology has benefited greatly from technological advances in equipment, computer programs and data bases, molecular biology, and immunochemistry. With respect to the detection of microorganisms in specimens, there has been a shift from the multistep methods previously discussed to unitary procedures and systems that incorporate standardization, speed, reproducibility, miniaturization, mechanization, and automation. These rapid identification methods can be divided into three categories: (1) manual biochemical systems, (2) mechanized/automated systems, and (3) immunologic systems.

One example of a “kit approach” biochemical system for the identification of members of the family *Enterobacteriaceae* and other gram-negative bacteria is the API 20E system. It consists of a plastic strip (figure 36.7) with 20 microtubes containing dehydrated biochemical substrates that can detect certain biochemical characteristics. The biochemical substrates in the 20 microtubes are inoculated with a pure culture of bacteria evenly suspended in sterile physiological saline. After 5 hours or overnight incubation, the 20 test results are converted to a seven- or nine-digit profile (figure 36.8). This profile number can be



Figure 36.7 A “Kit Approach” to Bacterial Identification. The API 20E manual biochemical system for microbial identification. (a) Positive and (b) negative results.

(a) Normal 7-digit code 5 144 572 = *E. coli*.



(b) 9-digit code 2 212 004 63 = *Pseudomonas aeruginosa*

Construction of a 9-digit profile

To the 7-digit profile illustrated in part a, 2 digits are added corresponding to the following characteristics:

NO ₂ : N ₂	Reduction of nitrate to nitrite only
GAS:	Complete reduction of nitrate to N ₂ gas or amines
MOT:	Observation of motility
MAC:	Growth on MacConkey medium
OF/O:	Oxidative utilization of glucose (OF-open)
OF/F:	Fermentative utilization of glucose (OF-closed)

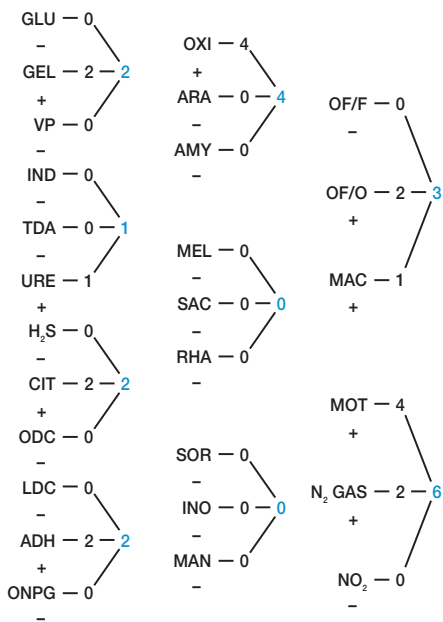


Figure 36.8 The API 20E Profile Number. The conversion of API 20E test results to the codes used in identification of unknown bacteria. The test results read top to bottom (and right to left in part b) correspond to the 7- and 9-digit codes when read in the right-to-left order. The tests required for obtaining a 7-digit code take an 18–24 hour incubation and will identify most members of the *Enterobacteriaceae*. The longer procedure that yields a 9-digit code is required to identify many gram-negative nonfermenting bacteria. The following tests are common to both procedures: ONPG (β-galactosidase); ADH (arginine dihydrolase); LDC (lysine decarboxylase); ODC (ornithine decarboxylase); CIT (citrate utilization); H₂S (hydrogen sulfide production); URE (urease); TDA (tryptophane deaminase); IND (indole production); VP (Voges-Proskauer test for acetoin); GEL (gelatin liquefaction); the fermentation of glucose (GLU), mannitol (MAN), inositol (INO), sorbitol (SOR), rhamnose (RHA), sucrose (SAC), melibiose (MEL), amygdalin (AMY), and arabinose (ARA); and OXI (oxidase test).

Table 36.3 Some Common Rapid Immunologic Test Kits for the Detection of Bacteria and Viruses in Clinical Specimens

Bactigen (Wampole Laboratories, Cranburg, N.J.) The Bactigen kit is used for the detection of <i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> type b, and <i>Neisseria meningitidis</i> groups A, B, C, and Y from cerebrospinal fluid, serum, and urine.
Culturette Group A Strep ID Kit (Marion Scientific, Kansas City, Mo.) The Culturette kit is used for the detection of group A streptococci from throat swabs.
Directigen (Hynson, Wescott, and Dunning, Baltimore, Md.) The Directigen Meningitis Test kit is used to detect <i>H. influenzae</i> type b, <i>S. pneumoniae</i> , and <i>N. meningitidis</i> groups A and C. The Directigen Group A Strep Test kit is used for the direct detection of group A streptococci from throat swabs.
Gono Gen (Micro-Media Systems, San Jose, Calif.) The Gono Gen kit detects <i>Neisseria gonorrhoeae</i> .
QuickVue <i>H. pylori</i> Test (Quidel, San Diego, Calif.) A seven minute test for detection of IgG antibodies against <i>Helicobacter pylori</i> in human serum or plasma.
Staphaurex (Wellcome Diagnostics, Research Triangle Park, N.C.) Staphaurex screens and confirms <i>Staphylococcus aureus</i> in 30 seconds.
Directigen RSV (Becton Dickinson Microbiology Systems, Cockeysville, Md.) By using a nasopharyngeal swab, the respiratory syncytial virus can be detected in 15 minutes.
SureCell Herpes (HSV) Test (Kodak, Rochester, N.Y.) Detects the herpes (HSV) 1 and 2 viruses in minutes.
SUDS HIV-1 Test (Murex Corporation, Norcross, Ga.) Detects antibodies to HIV-1 antigens in about 10 minutes.

used with a computer or a book called the *API Profile Index* to find the name of the bacterium.

Immunologic Techniques

The culturing of certain viruses, bacteria, fungi, and parasites from clinical specimens may not be possible because the methodology remains undeveloped (*Treponema pallidum*; hepatitis A, B, C; and Epstein-Barr virus), is unsafe (rickettsias), or is impractical for all but a few clinical microbiology laboratories (mycobacteria). Cultures also may be negative because of prior antimicrobial therapy. Under these circumstances, detection of antibodies or antigens may be quite valuable diagnostically.

Immunologic systems for the detection and identification of pathogens from clinical specimens are easy to use, give relatively rapid reaction endpoints, and are sensitive and specific (they give a low percentage of false positives and negatives). Some of the more popular immunologic rapid test kits for viruses and bacteria are presented in **table 36.3**.

Each individual's immunologic response to a microorganism is quite variable. As a result the interpretation of immunologic tests is sometimes difficult. For example, a single elevated antibody IgM titer usually does not distinguish between active and past infections. Furthermore, the lack of a measurable antibody titer may reflect either a microorganism's lack of immunogenicity or an insufficient time for an antibody response to develop following the onset of the infectious disease. Some pa-

tients also are immunosuppressed due to other disease processes and/or treatment procedures (e.g., cancer and AIDS patients) and therefore do not respond. For these reasons, test selection and timing of specimen collection are essential to the proper interpretation of immunologic tests. [Antibody titer](#) (pp. 742, 776).

The most widely used immunologic techniques available to detect microorganisms in clinical specimens are covered in detail in section 33.3. No single technique is universally applicable for measuring an individual's immunologic response to all microorganisms. Techniques are therefore chosen based on their selectivity, specificity, ease, speed of performance, and cost-effectiveness.

1. Describe in general how biochemical tests are used in the API 20E system to identify bacteria.
2. Name the two basic immunologic procedures used in kits to identify microorganisms.
3. Why might cultures for some microorganisms be unavailable?
4. Why are test selection and timing of specimen collection essential to the proper interpretation of immunologic tests?

Bacteriophage Typing

Bacteriophages (phages) are viruses that attack members of a particular bacterial species, or strains within a species (*see chapter 17*). **Bacteriophage (phage) typing** is based on the specificity of phage surface receptors for cell surface receptors. Only those bacteriophages that can attach to these surface receptors can infect bacteria and cause lysis. On a petri dish culture, lytic bacteriophages cause plaques on lawns of sensitive bacteria. These plaques represent infection by the virus (*see figure 16.4*).

In bacteriophage typing the clinical microbiologist inoculates the bacterium to be tested onto a petri plate. The plate is heavily and uniformly inoculated with a cotton swab so that the bacteria will grow to form a solid sheet or lawn of cells. No uninoculated areas should be left. The plate is then marked off into squares (15 to 20 mm per side), and each square is inoculated with a drop of suspension from the different phages available for typing. After the plate is incubated for 24 hours, it is observed for plaques. The phage type is reported as a specific genus and species followed by the types that can infect the bacterium. For example, the series 10/16/24 indicates that this bacterium is sensitive to phages 10, 16, and 24, and belongs to a collection of strains, called a **phagovar**, that have this particular phage sensitivity. Bacteriophage typing remains a tool of the research and reference laboratory.

Molecular Methods and Analysis of Metabolic Products

With the application of new molecular technology, it is now possible to analyze the molecular characteristics of microorganisms in the clinical laboratory. Some of the most accurate approaches to microbial identification are through the analysis of proteins and nucleic acids. Examples previously discussed (*see chapter 19*) include comparison of proteins; physical, kinetic, and regulatory properties of microbial enzymes; nucleic acid-base composition (*see table 19.5*); nucleic acid hybridization; and nucleic acid sequencing.

Three other molecular methods being widely used are nucleic acid probes, gas-liquid chromatography, and plasmid fingerprinting.

Nucleic Acid-Based Detection Methods

A recent development in clinical microbiology is the use of nucleic acid-based diagnostic methods for the detection and identification of microorganisms. For example, DNA probe technology (*see section 14.4*) identifies a microorganism by probing its genetic composition. The use of cloned DNA as a probe is based upon the capacity of single-stranded DNA to bind (hybridize) with a complementary nucleic acid sequence present in test specimens to form a double-stranded DNA hybrid (**figure 36.9**). Thus a single-stranded sequence derived from one microorganism (the probe) is used to search for others containing the same sequence. This hybridization reaction may be applied to purified DNA preparations, to bacterial colonies, or to clinical specimens such as tissue, serum, sputum, and pus. Recently DNA probes have been developed that bind to complementary strands of ribosomal RNA. These DNA:rRNA hybrids are more sensitive than conventional DNA probes, give results in 2 hours or less, and require the presence of fewer microorganisms. DNA probe sensitivity can be increased by over one million-fold if the target DNA is first amplified using the polymerase chain reaction (*see section 14.3*). DNA:rRNA probes are available or are currently being developed for most clinically important microorganisms.

Ribosomal RNA from *E. coli* can be used to type bacterial strains by probing chromosomal DNA in Southern blots (*see figure 14.5*). This method of strain typing, called **ribotyping**, makes use of the fact that rRNA genes are scattered throughout the chromosome of most bacteria. When the chromosomes of several strains are cleaved using restriction endonucleases and the digests are analyzed with the Southern blotting procedure, rRNA probes will produce different patterns with different strains. Cloned rRNA genes also can be used as probes instead of *E. coli* rRNA, and similar banding patterns result.

Gas-Liquid Chromatography

During chromatography a chemical mixture carried by a liquid or gas is separated into its individual components because of processes such as adsorption, ion-exchange, and partitioning between different solvent phases. In gas-liquid chromatography (GLC), specific microbial metabolites, cellular fatty acids, and products from the pyrolysis (a chemical change caused by heat) of whole bacterial cells are analyzed and identified. These compounds are easily removed from growth media by extraction with an organic solvent such as ether. The ether extract is then injected into the GLC system. Both volatile and nonvolatile acids can be identified. Based on the pattern of fatty acid production, common bacteria isolated from clinical specimens can be identified.

The reliability, precision, and accuracy of GLC have been improved significantly with continued advances in instrumentation; the introduction of instruments for high-performance liquid chromatography; and the use of mass spectrometry, nuclear magnetic resonance spectroscopy, and associated analytical techniques for the identification of components separated by the chro-

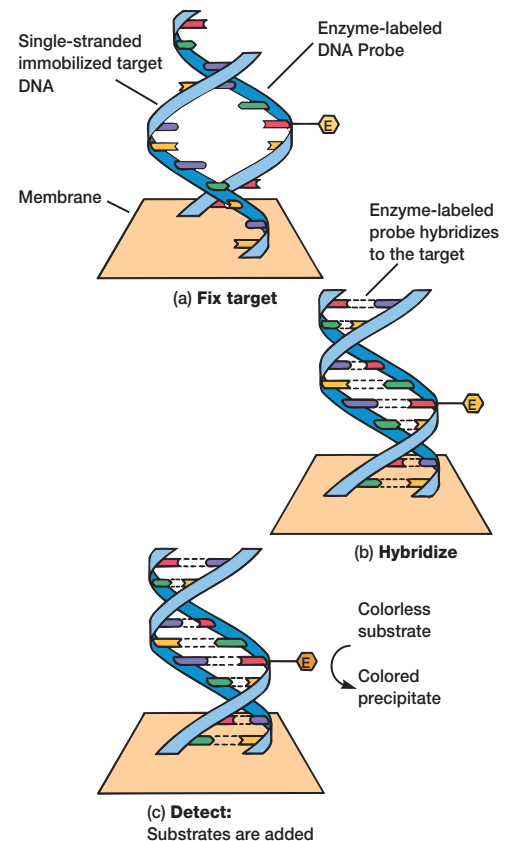


Figure 36.9 Basic Steps in A DNA Probe Hybridization Assay.

(a) Single-stranded target nucleic acid is bound to a membrane. A DNA probe with attached enzyme (E) also is employed. (b) The probe is added to the membrane. If the probe hybridizes to the target DNA, a double-stranded DNA hybrid is formed. (c) A colorless substrate is added. The enzyme attached to the probe converts the substrate to a colored precipitate. This detection system is semiquantitative, in that color intensity is proportional to the quantity of hybridized target nucleic acid present.

matographic process. These combined techniques have recently been used to discover specific chemical markers of various infectious disease agents by direct analysis of body fluids.

Plasmid Fingerprinting

As presented in section 13.2, a plasmid is an autonomously replicating extrachromosomal molecule of DNA in bacteria. **Plasmid fingerprinting** identifies microbial isolates of the same or similar strains; related strains often contain the same number of plasmids with the same molecular weights and similar phenotypes. In contrast, microbial isolates that are phenotypically distinct have different plasmid fingerprints. Plasmid fingerprinting of many *E. coli*, *Salmonella*, *Campylobacter*, and *Pseudomonas* strains and species has demonstrated that this method often is more accurate

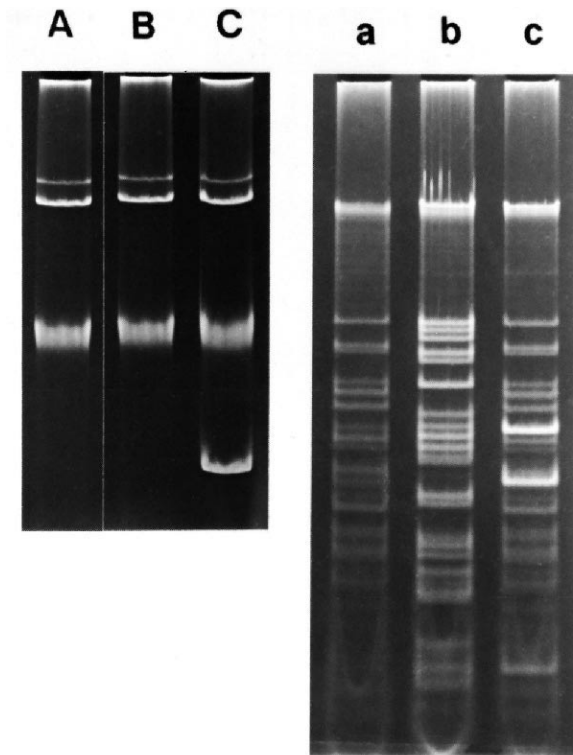


Figure 36.10 Plasmid Fingerprinting. Agarose gel electrophoresis of plasmid DNA.

than other phenotyping methods such as biotyping, antibiotic resistance patterns, phage typing, and serotyping.

The technique of plasmid fingerprinting involves five steps:

1. The bacterial strains are grown in broth or on agar plates.
2. The cells are harvested and lysed with a detergent.
3. The plasmid DNA is separated from the chromosomal DNA.
4. The plasmid DNA is applied to agarose gels and electrophoretically separated.
5. The gel is stained with ethidium bromide, which binds to DNA, causing it to fluoresce under UV light. The plasmid DNA bands are then located.

Because the migration rate of plasmid DNA in agarose is inversely proportional to the molecular weight, plasmids of a different size appear as distinct bands in the stained gel. The molecular weight of each plasmid species can then be determined from a plot of the distance that each species has migrated versus the log of the molecular weights of plasmid markers of known size that have been electrophoresed simultaneously in the same gel (figure 36.10).

36.3 Susceptibility Testing

Many clinical microbiologists believe that determining the susceptibility of a microorganism to specific antibiotics is one of the most important tests performed in the clinical microbiology lab-

oratory. Results (figure 36.5o) can show the antibiotics to which a microorganism is most susceptible and the proper therapeutic dose needed to treat the infectious disease. (Dilution susceptibility tests, disk diffusion tests [Kirby-Bauer method], and drug concentration measurements in the blood are discussed in detail in section 35.3.)

1. What is the basis for bacteriophage typing?
2. How can nucleic acid-based detection methods be used by the clinical microbiologist? Gas-liquid chromatography?
3. How can a suspect bacterium be plasmid fingerprinted? Why is susceptibility testing so important in clinical microbiology?

36.4 Computers in Clinical Microbiology

Computer systems in the clinical microbiology laboratory are designed primarily to replace the handwritten mode of information acquisition and transmission. Computers improve the efficiency of the laboratory operation and increase the speed and clarity with which results can be reported to physicians. From a work-flow standpoint, the major functions involving the computer are test ordering, result entry, analysis of results, and report preparation (figure 36.1h, i).

Test orders may be entered into the computer from the hospital unit or laboratory. Standard order practices should include specific requests (e.g., rule out *Nocardia* and diphtheria), all pertinent patient data, and an accession number. Once the test order has been placed, the system should allow the usual work flow to proceed with the labeled clinical specimen, date, test order, and computer accession number.

After clinical results are obtained in the laboratory, they are entered into a written log and then into the computer. To meet the many needs of microbiological entry, the computer system must be rapid and flexible in its entry modes.

Printed reports of the patient's laboratory findings are the product of the computer system. Print programs also should permit flexible formatting of the reports so that additional data can be generated. For example, the computer system should be able to generate cumulative reports that summarize days to weeks of an inpatient stay.

Besides reporting laboratory tests, computers manage specimen logs, reports of overdue tests, quality control statistics, antimicrobial susceptibility probabilities, hospital epidemiological data, and many other items. The computer can be interfaced with various automated instruments for rapid and accurate calculation and transfer of clinical data.

1. What are some different ways in which computers can be used in the clinical microbiology laboratory?
2. From the standpoint of work flow, how can computers be specifically used in a clinical microbiology laboratory?

Summary

1. The major focus of the clinical microbiologist is to isolate and identify microorganisms from clinical specimens accurately and rapidly (**figure 36.1**). A clinical specimen represents a portion or quantity of biological material that is tested, examined, or studied to determine the presence or absence of specific microorganisms.
2. Specimens may be collected by various methods (**figure 36.2**) that include swabs, needle aspiration, intubation, catheters, and clean-catch techniques. Each method is designed to ensure that only the proper material will be sent to the clinical laboratory.
3. Immediately after collection the specimen must be properly handled and labeled. Speed in transporting the specimen to the clinical laboratory after it has been collected is of prime importance.
4. The clinical microbiology laboratory can provide preliminary or definitive identification of microorganisms based on (a) microscopic examination of specimens; (b) growth and biochemical characteristics of microorganisms isolated from cultures (**figure 36.5**); and (c) immunologic techniques that detect antibodies or microbial antigens.
5. Viruses are identified by isolation in living cells or immunologic tests. Several types of living cells are available: cell culture, embryonated hen's eggs, and experimental animals. Rickettsial disease can be diagnosed immunologically or by isolation of the organism. Chlamydiae can be demonstrated in tissue and cell scrapings with Giemsa stain, which detects the characteristic intracellular inclusion bodies. The most routinely used techniques for identification of the mycoplasmas are immunologic. Identification of fungi often can be made if a portion of the specimen is mixed with a drop of 10% Calcofluor White stain. Wet mounts of stool specimens or urine can be examined microscopically for the presence of parasites.
6. The initial identity of a bacterial organism may be suggested by (1) the source of the culture specimen; (2) its microscopic appearance; (3) its pattern of growth on selective, differential, enrichment, or characteristic media; and (4) its hemolytic, metabolic, and fermentative properties.
7. Rapid methods for microbial identification can be divided into three categories: (1) manual biochemical systems (**figure 36.7**), (2) mechanized/automated systems, and (3) immunologic systems.
8. Bacteriophage typing for bacterial identification is based on the fact that phage surface receptors bind to specific cell surface receptors. On a petri plate culture, bacteriophages cause plaques on lawns of bacteria with the proper receptors.
9. Various molecular methods and analyses of metabolic products also can be used to identify microorganisms. Examples include nucleic acid-based detection, gas-liquid chromatography, and plasmid fingerprinting.
10. After the microorganism has been isolated, cultured, and/or identified, samples are used in susceptibility tests to find which method of control will be most effective. The results are provided to the physician as quickly as possible.
11. Computer systems in clinical microbiology are designed to replace handwritten information exchange and to speed data evaluation and report preparation.

Key Terms

bacteriophage (phage) typing 842
catheter 827
clinical microbiologist 827
cytopathic effect 832

hemadsorption 832
intubation 827
needle aspiration 827
phagovar 842

plasmid fingerprinting 843
ribotyping 843
sputum 829
swab 827

Questions for Thought and Review

1. How can clinical specimens be taken from a patient with various infectious diseases? Give specific examples of procedures used.
2. What precaution must be observed when a culture is obtained from the respiratory system?
3. Why is gas-liquid chromatography a useful approach to the identification of anaerobes?
4. How does a clinical microbiologist convert an API 20E test result to a numerical code for bacterial identification?
5. How is a dichotomous key used in bacterial identification?
6. What are some different ways in which biochemical reactions can be used to identify microorganisms?
7. What are some advantages of automation in the clinical microbiology laboratory?
8. When should laboratory animals be used in the identification of microorganisms?
9. Why is plasmid fingerprinting such an accurate method for the identification of microorganisms?

Critical Thinking Questions

1. As more new ways of identifying the characteristics of microorganisms emerge, the number of distinguishable microbial strains also seems to increase. Why do you think this occurs?
2. Why are miniaturized identification systems used in clinical microbiology? Describe one such system and its advantage over classic dichotomous keys.

Additional Reading

General

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